Faculty of Engineering & Technology

Electrical & Computer Engineering Department

ENCS3320, Computer Networks

Project 2 – Cisco Packet Tracer

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Abstract

In this project, we built a small simulated network using Cisco Packet Tracer. We started by dividing a large IP address block into smaller subnets to fit different areas of the network, including the core, university, street, home, and data center. We connected routers, switches, PCs, phones, and servers, and used OSPF to allow communication between all parts. We also configured DHCP to give IP addresses automatically, and added DNS, web, and mail servers in the data center so users could browse a webpage and send emails. Wireless access was set up for the university and mobile users through a 4G cell tower. Finally, we tested the network using tools like ping and traceroute, and confirmed mostly worked. This project helped us practice important networking skills like subnetting, routing, server setup, and teamwork.

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Theory and procedure

This project focuses on building a basic computer network that connects multiple areas such as a university, data center, street, and home.

1. IP Addressing and Subnetting

To manage the IP addresses efficiently, we used subnetting. This means dividing one large IP block into smaller subnetworks (subnets) so that each part of the network gets just the number of IP addresses it needs. Subnetting helps reduce waste of addresses and improves routing efficiency. We followed the rules of Classless Inter-Domain Routing (CIDR) and calculated the number of hosts per subnet to assign the right subnet mask.

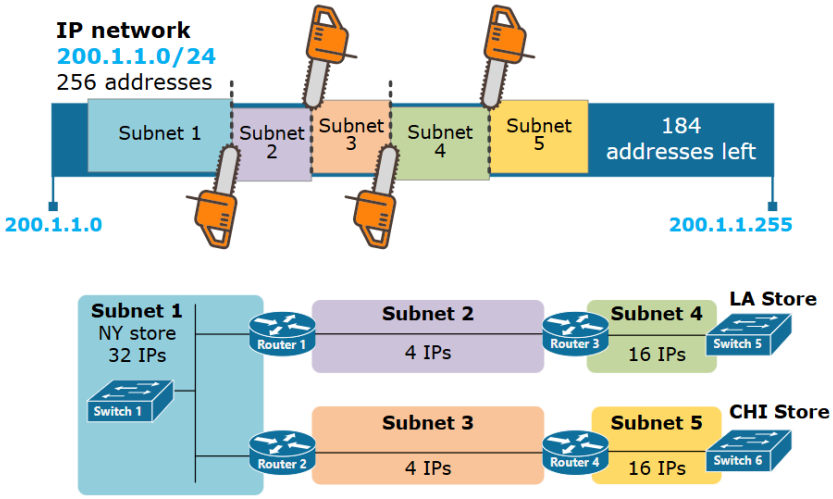


Figure 1: Subnetting example.[1]

2. Routing and OSPF Protocol

To connect all parts of the network and let data move between them, we used routers and the OSPF (Open Shortest Path First) protocol. OSPF is a dynamic and smart routing protocol that finds the best path for data by using the shortest route. If one path goes down, OSPF quickly finds another path to keep the network working. It is a link-state protocol, which means each router shares information about the links it's directly connected to. This allows all routers in the same area to have a full map of the network and calculate the best way to reach every destination. [3]

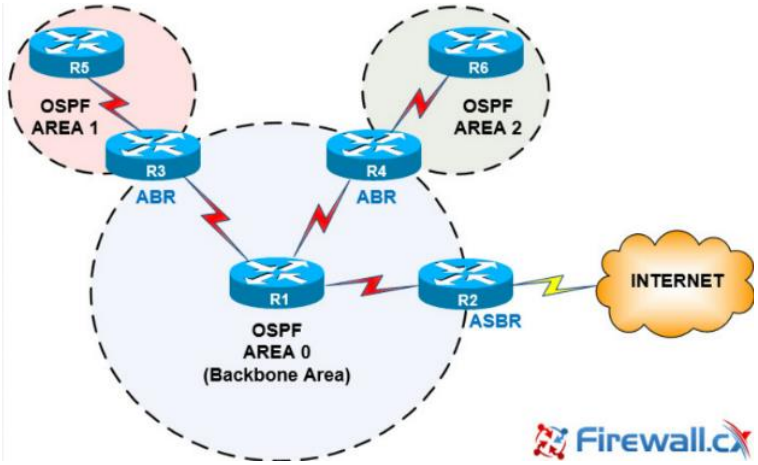


Figure 2: OSPF protocol.[2]

3. DHCP (Dynamic Host Configuration Protocol)

The Dynamic Host Configuration Protocol (DHCP) is a client-server protocol that automatically gives devices on the network their IP address, along with other important settings like the subnet mask and default gateway.[4]

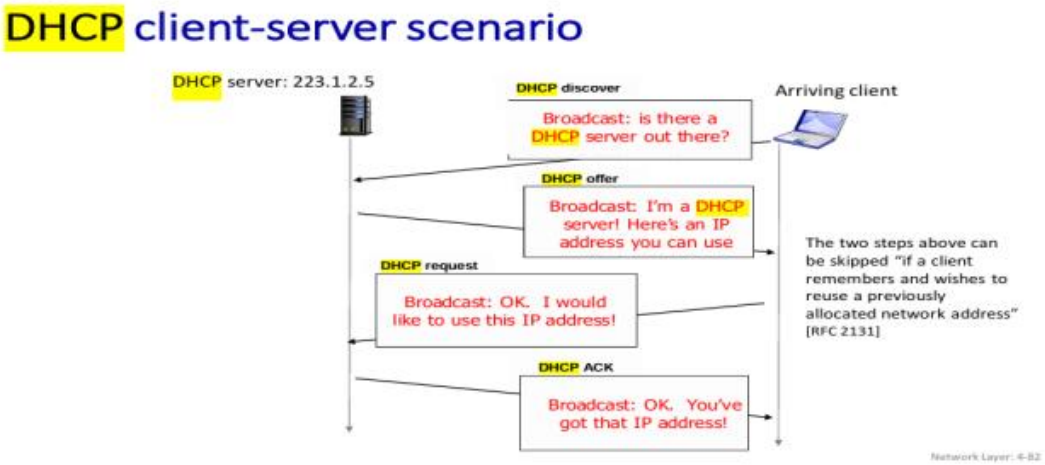


Figure 3: Dynamic Host Configuration Protocol (client-server scenario).

Advantages of DHCP:

- **Automatic IP Assignment:** Simplifies network administration by automatically assigning IP addresses to devices, reducing the need for manual configuration.
- **Efficient IP Utilization:** Ensures IP addresses are distributed effectively, preventing address conflicts and maximizing the use of the available IP pool.
- **Scalable Network Integration:** Facilitates the seamless addition of new devices to the network, supporting expansion without manual intervention.

Disadvantages of DHCP:

- **Security Vulnerabilities:** If not properly secured, DHCP can be exploited by unauthorized devices or rogue servers, potentially leading to network issues.
- **Dependency on DHCP Server:** Devices rely on the DHCP server to obtain or renew IP addresses; if the server fails, network connectivity may be interrupted.

4. DNS, Web, and Mail Services

The data centre hosts three important services:

- **DNS (Domain Name System):** DNS is a hierarchical and distributed naming system that translates domain names into IP addresses.[6]

- **Web Server:** A web server is a system—comprising software, hardware, or a combination of both—that stores, processes, and serves web content to users via the Internet using the HTTP or HTTPS protocols.[7]

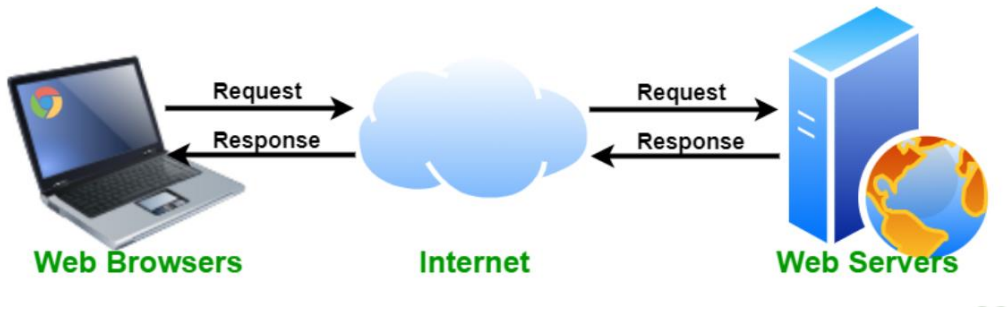


Figure 4: Web server.[7]

- **Mail Server:** Allows users to send and receive emails using protocols like SMTP and POP3.

These services make the network more useful and realistic, just like in real-world IT environments.

5. Wireless and Mobile Networking

"Wireless" refers to communication without the use of physical wires, relying instead on electromagnetic waves or infrared signals. All wireless devices include antennas or sensors to send and receive these signals. Common examples of wireless devices include mobile phones, wireless sensors, TV remotes, satellite dish receivers, and laptops equipped with WLAN cards. In wireless networks, data or voice communication is carried out through radio frequency waves instead of wired connections.[8]

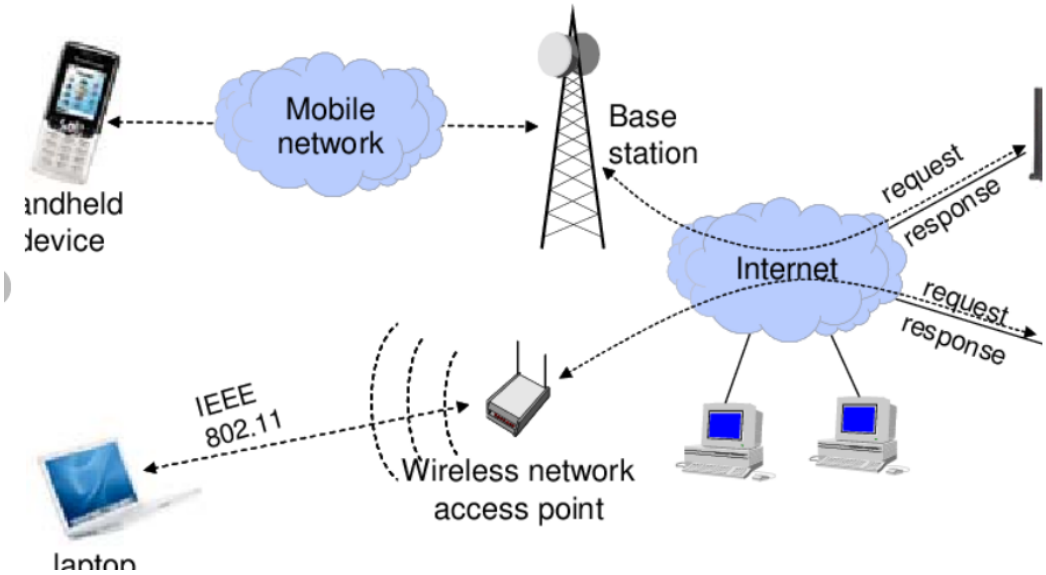


Figure 5: Wireless and mobile networking. [9]

Results and Discussion

Our Topology

The network topology consists of five interconnected areas: the Core network (Area 0), University network (Area 1), Street network (Area 2), Home network (Area 3), and Datacenter network (Area 4), all linked through the Core network as shown in Figure 6.

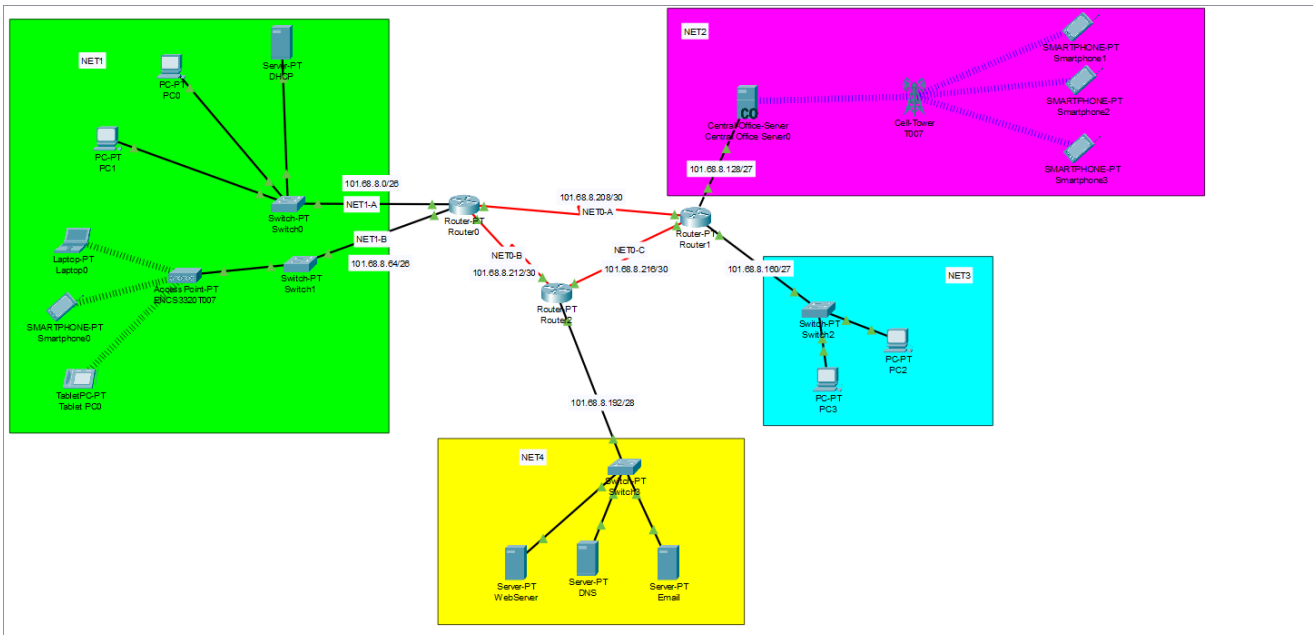


Figure 6: The Full Topology.

Tasks

1. **<u>IP subnetting task:</u>**

   To perform the subnetting task, we began with the main IP block 1X.Y.8.0/23, where X and Y are the last four digits of the team leader's student ID (SID). Since the team leader's SID (Al_Zahra Nassif) is 1220168, we used X = 01 and Y = 68, resulting in the base network IP of 101.68.8.0/23. This provides 512 IP addresses, which are then divided into smaller subnets according to the number of hosts required for each subnetwork.

Network	Subnetwork	Hosts	Host Bits Needed
Core	NET0-A	2	2
Core	NET0-B	2	2
Core	NET0-C	2	2
University	NET1-A	60	6
University	NET1-B	60	6
Street	NET2	30	5
Home	NET3	20	5
Datacenter	NET4	15	5

Table 1: Area's Hosts.

Subnetwork	Network Address	CIDR	Subnet Mask	Usable IP Range	Broadcast IP
NET1-A	101.68.8.0	/26	255.255.255.192	101.68.8.1 – 101.68.8.62	101.68.8.63
NET1-B	101.68.8.64	/26	255.255.255.192	101.68.8.65 – 101.68.8.126	101.68.8.127
NET2	101.68.8.128	/27	255.255.255.224	101.68.8.129 – 101.68.8.158	101.68.8.159
NET3	101.68.8.160	/27	255.255.255.224	101.68.8.161 – 101.68.8.190	101.68.8.191
NET4	101.68.8.192	/28	255.255.255.240	101.68.8.193 – 101.68.8.206	101.68.8.207
NET0-A	101.68.8.208	/30	255.255.255.252	101.68.8.209 – 101.68.8.210	101.68.8.211
NET0-B	101.68.8.212	/30	255.255.255.252	101.68.8.213 – 101.68.8.214	101.68.8.215
NET0-C	101.68.8.216	/30	255.255.255.252	101.68.8.217 – 101.68.8.218	101.68.8.219

Table 2: IP Subnetting.

2. **<u>Building Topology Task:</u>**

- **Core network (Area 0):**

We create a core network (Area0), the network contains three routers, as the figure below:

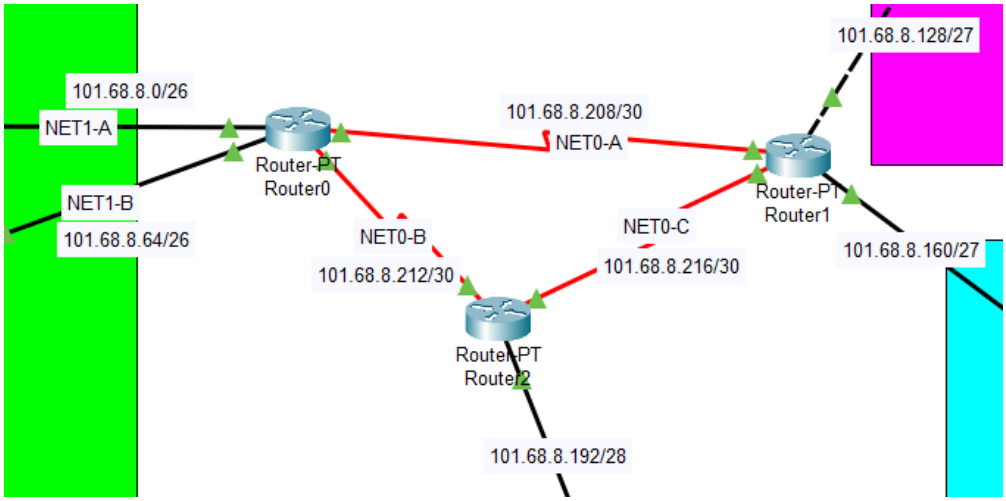


Figure 7: Core network (Area 0).

First of all, we assigned a static IP configurations to all router ports, as illustrate in the following sections.

1) **IP Configuration for Router0:**

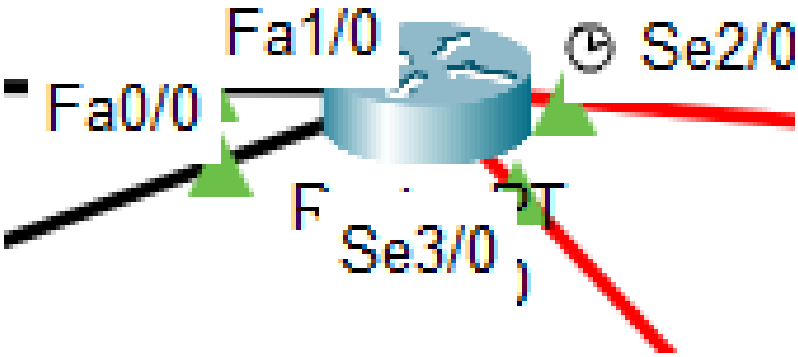


Figure 8: Router0 ports.

As we show in the figure above, Router0 has four connected ports, which are:

- Fa0/0 port connected with network 101.68.8.64/26, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.65.

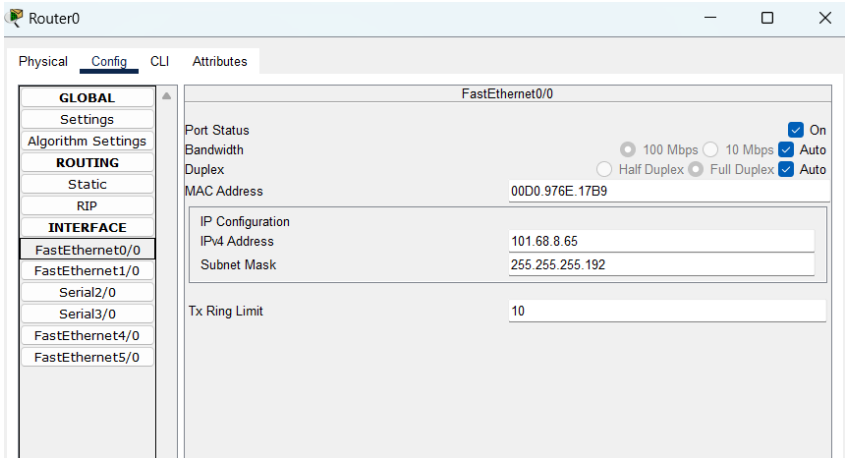


Figure 9: IP Configuration for Router0 Port Fa0/0.

- Fa1/0 port connected with network 101.68.8.0/26, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.1

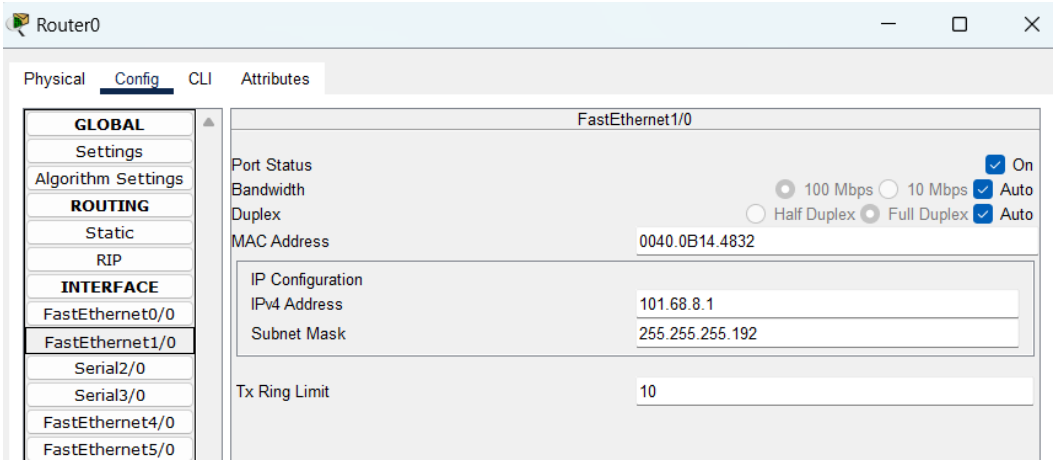


Figure 10: IP Configuration for Router0 Port Fa1/0.

- Se2/0 port connected with network 101.68.8.208/30, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.209

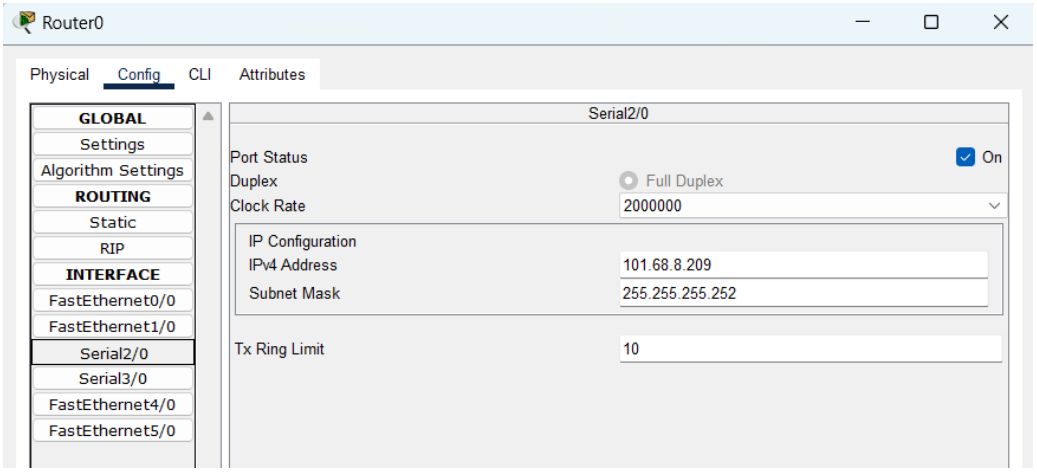


Figure 11: IP Configuration for Router0 Port Se2/0.

- Se3/0 port connected with network 101.68.8.212/30, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.213

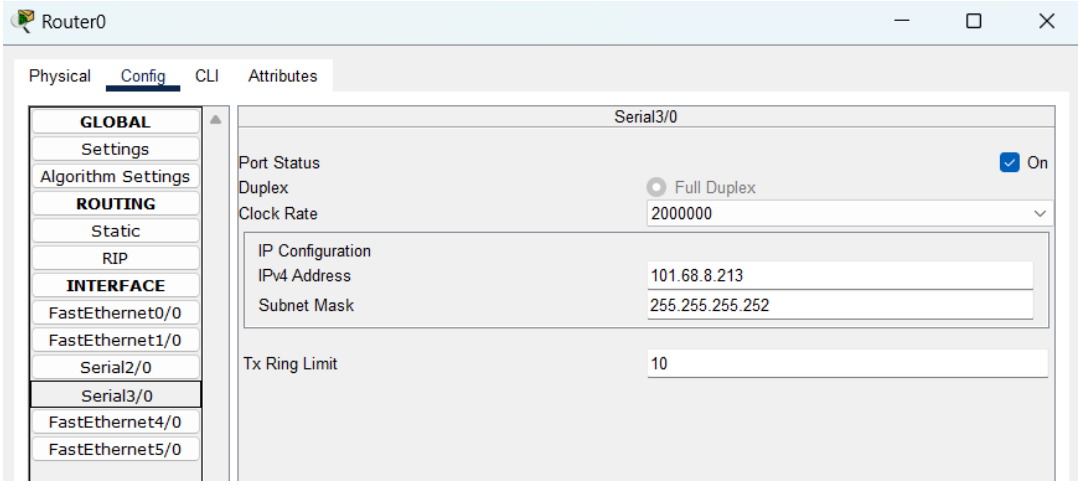


Figure 12: IP Configuration for Router0 Port Se3/0.

2) IP Configuration for Router1:

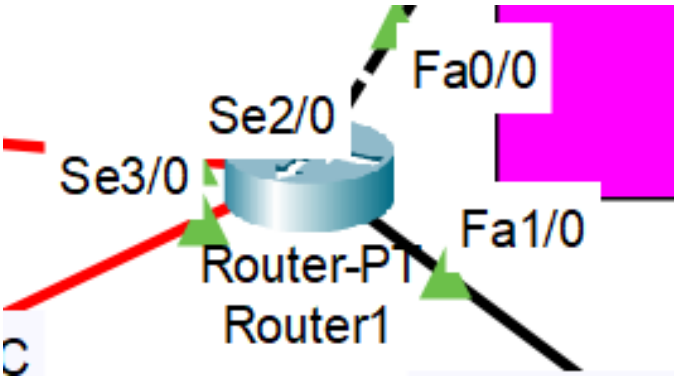


Figure 13: Router1 ports.

As we show in the figure above, Router1 has four connected ports, which are:

- Fa0/0 port connected with network 101.68.8.128/27, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.129

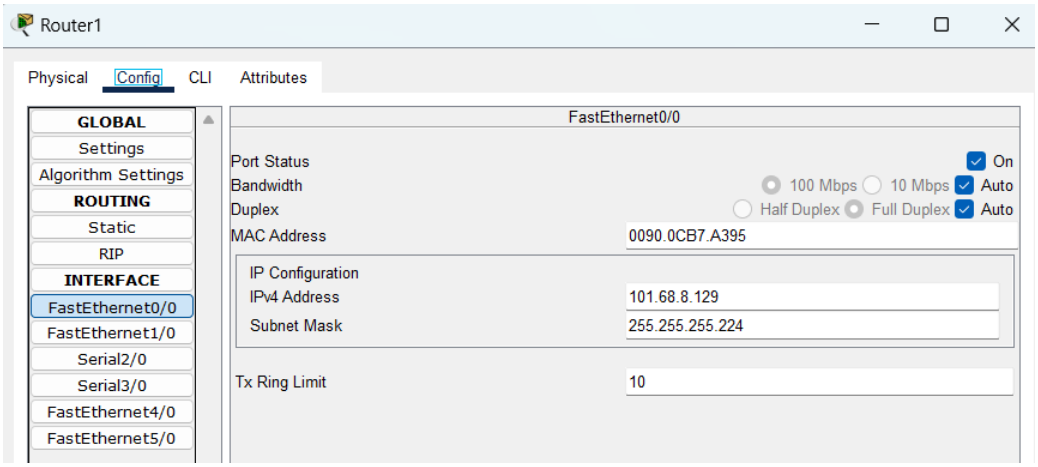


Figure 14: IP Configuration for Router1 Port Fa0/0.

- Fa1/0 port connected with network 101.68.8.160/27, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.161

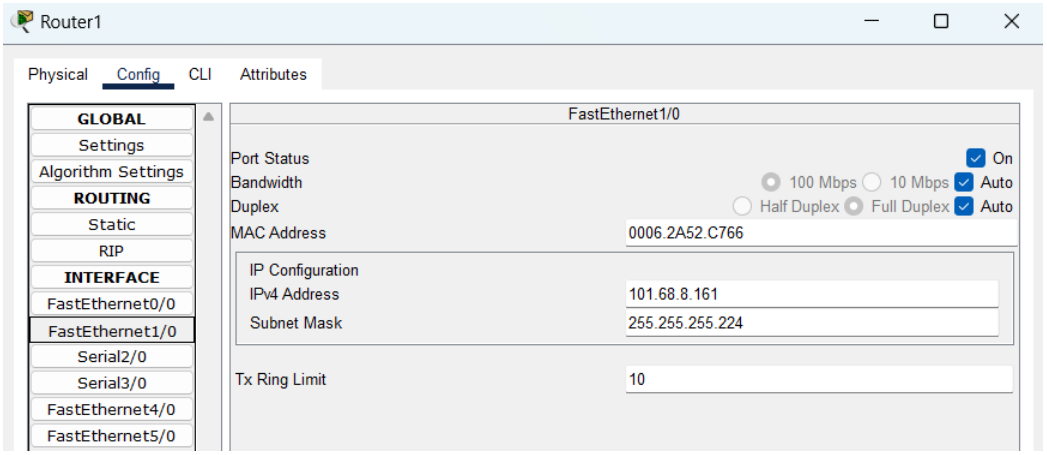


Figure 15: IP Configuration for Router1 Port Fa1/0.

- Se2/0 port connected with network 101.68.8.208/30, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.210

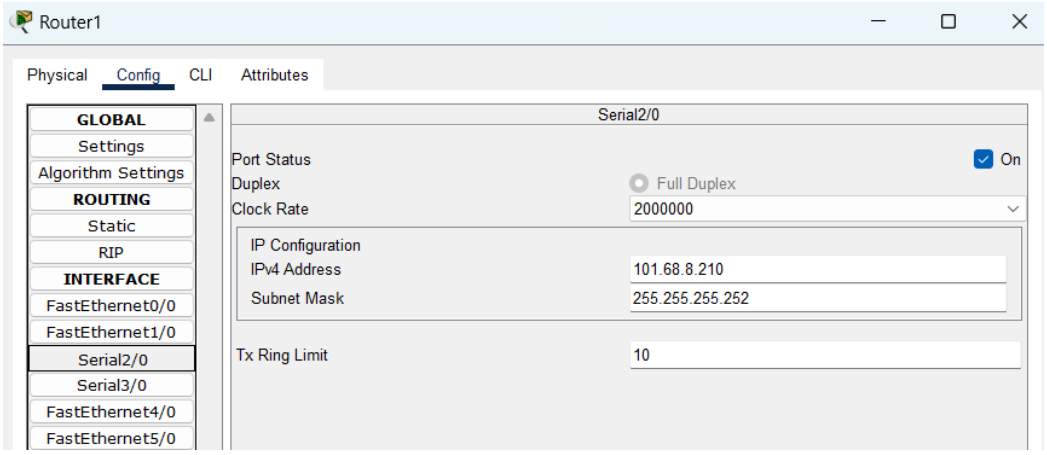


Figure 16: IP Configuration for Router1 Port Se2/0.

- Se3/0 port connected with network 101.68.8.216/30, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.218

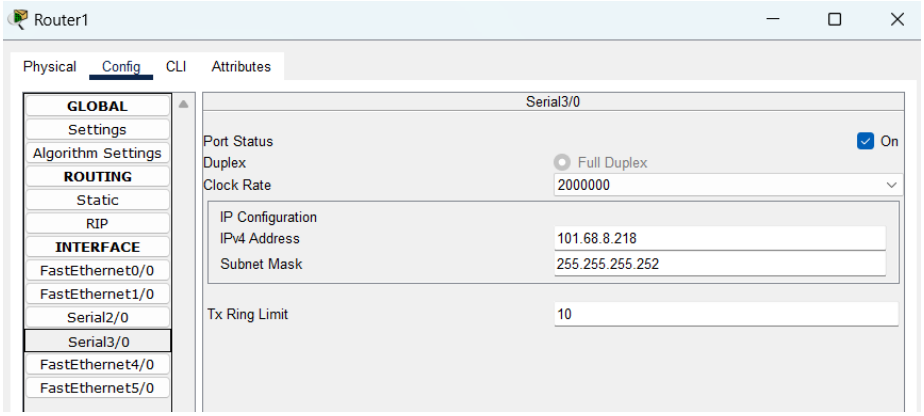


Figure 17: IP Configuration for Router1 Port Se3/0.

3) IP Configuration for Router2:

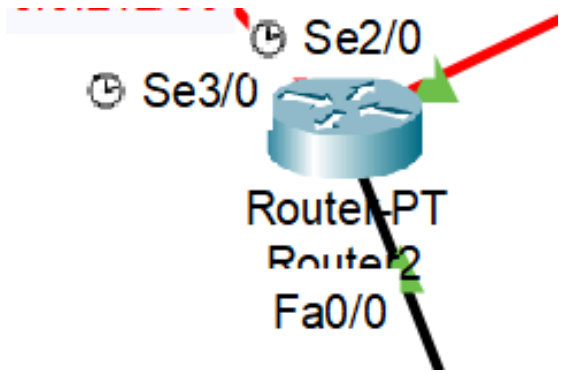


Figure 18: Router2 ports.

As we show in the figure above, Router1 has three connected ports, which are:

- Fa0/0 port connected with network 101.68.8.192/28, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.193

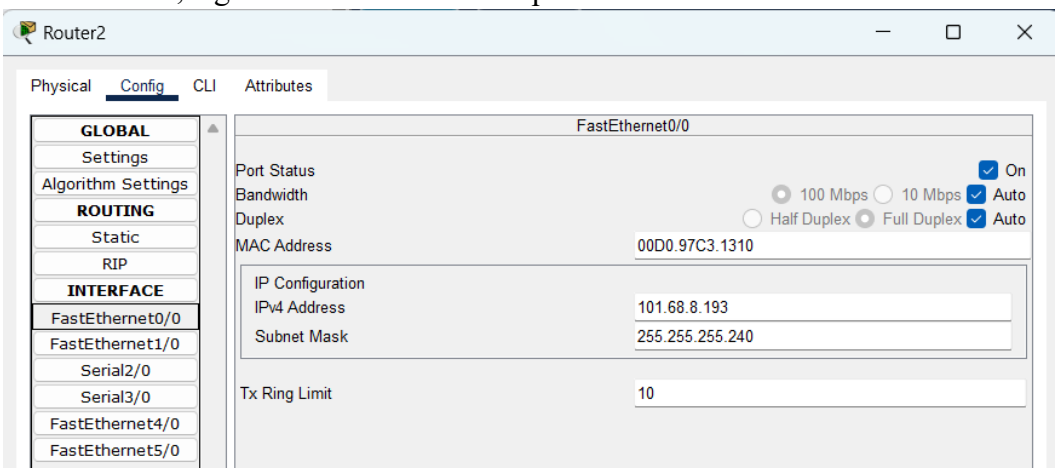


Figure 19: IP Configuration for Router1 Port Fa0/0.

- Se2/0 port connected with network 101.68.8.216/30, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.217

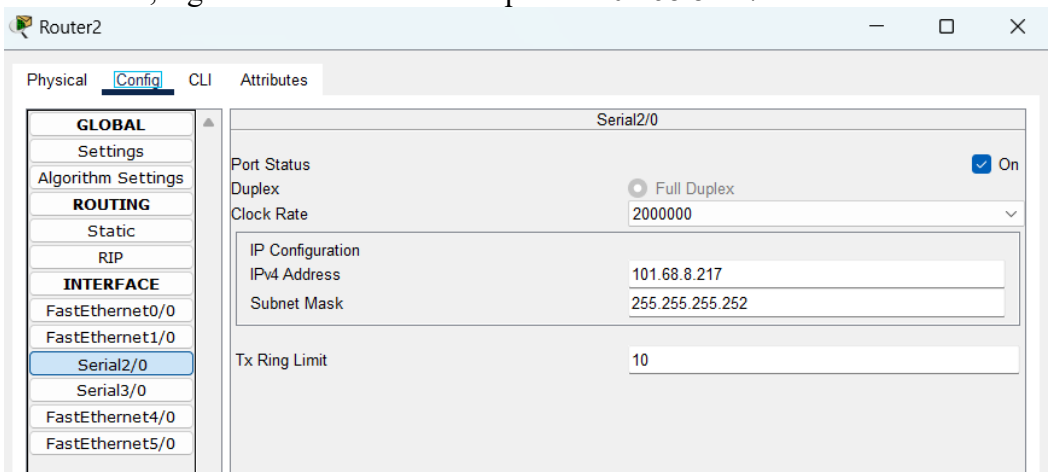


Figure 20: IP Configuration for Router1 Port Se2/0.

- Se3/0 port connected with network 101.68.8.212/30, so allocate a static IP address for this port from this network, e.g. the IP address for this port is 101.68.8.214

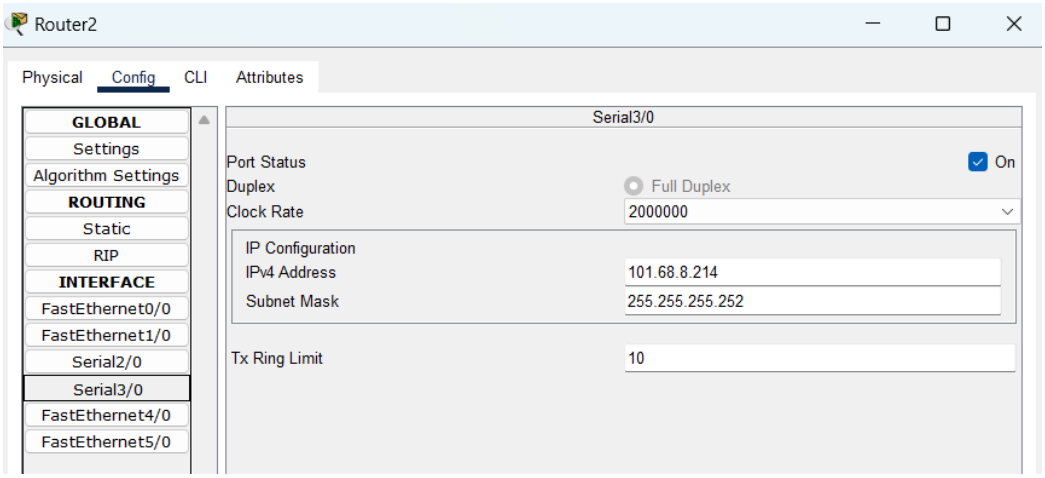


Figure 21: IP Configuration for Router1 Port Se3/0.

❖ **<u>Routing Configuration:</u>**

We used OSPF with process ID 1 on all routers to connect the five areas. Router0 (core) is in Area 0 and connects to Router1, Router2, and the rest of the network using /30 subnets (NET0-A, B, C). Router0 is connected to the University network (NET1-A, NET1-B), Router1 connects to the Street network (NET2) and to the Home network (NET3), and Route2 connects to the Datacenter network (NET4). Each router advertises its local networks and learns the rest through OSPF. The routing tables show a mix of directly connected (C) and OSPF-learned routes (O and O IA), proving that OSPF is working correctly and all routers are exchanging routes across areas.

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     101.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C       101.68.8.0/26 is directly connected, FastEthernet1/0
C       101.68.8.64/26 is directly connected, FastEthernet0/0
O IA    101.68.8.128/27 [110/65] via 101.68.8.210, 00:07:14, Serial2/0
O IA    101.68.8.160/27 [110/65] via 101.68.8.210, 00:07:14, Serial2/0
O IA    101.68.8.192/28 [110/65] via 101.68.8.214, 00:07:14, Serial3/0
C       101.68.8.208/30 is directly connected, Serial2/0
C       101.68.8.212/30 is directly connected, Serial3/0
O       101.68.8.216/30 [110/128] via 101.68.8.210, 00:07:14, Serial2/0
                        [110/128] via 101.68.8.214, 00:07:14, Serial3/0
```

Figure 22: Router0 - Routing configuration.

```
     101.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
O IA    101.68.8.0/26 [110/65] via 101.68.8.209, 00:21:29, Serial2/0
O IA    101.68.8.64/26 [110/65] via 101.68.8.209, 00:21:29, Serial2/0
C       101.68.8.128/27 is directly connected, FastEthernet0/0
C       101.68.8.160/27 is directly connected, FastEthernet1/0
O IA    101.68.8.192/28 [110/65] via 101.68.8.217, 00:21:29, Serial3/0
C       101.68.8.208/30 is directly connected, Serial2/0
O       101.68.8.212/30 [110/128] via 101.68.8.209, 00:21:29, Serial2/0
                        [110/128] via 101.68.8.217, 00:21:29, Serial3/0
C       101.68.8.216/30 is directly connected, Serial3/0

Router#
```

Figure 23: Router1 - Routing configuration.

```
     101.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
O IA    101.68.8.0/26 [110/65] via 101.68.8.213, 00:23:53, Serial3/0
O IA    101.68.8.64/26 [110/65] via 101.68.8.213, 00:23:53, Serial3/0
O IA    101.68.8.128/27 [110/65] via 101.68.8.218, 00:23:43, Serial2/0
O IA    101.68.8.160/27 [110/65] via 101.68.8.218, 00:23:43, Serial2/0
C       101.68.8.192/28 is directly connected, FastEthernet0/0
O       101.68.8.208/30 [110/128] via 101.68.8.218, 00:23:43, Serial2/0
                        [110/128] via 101.68.8.213, 00:23:43, Serial3/0
C       101.68.8.212/30 is directly connected, Serial3/0
C       101.68.8.216/30 is directly connected, Serial2/0
```

Figure 24: Router2 - Routing configuration.

University network (Area 1):

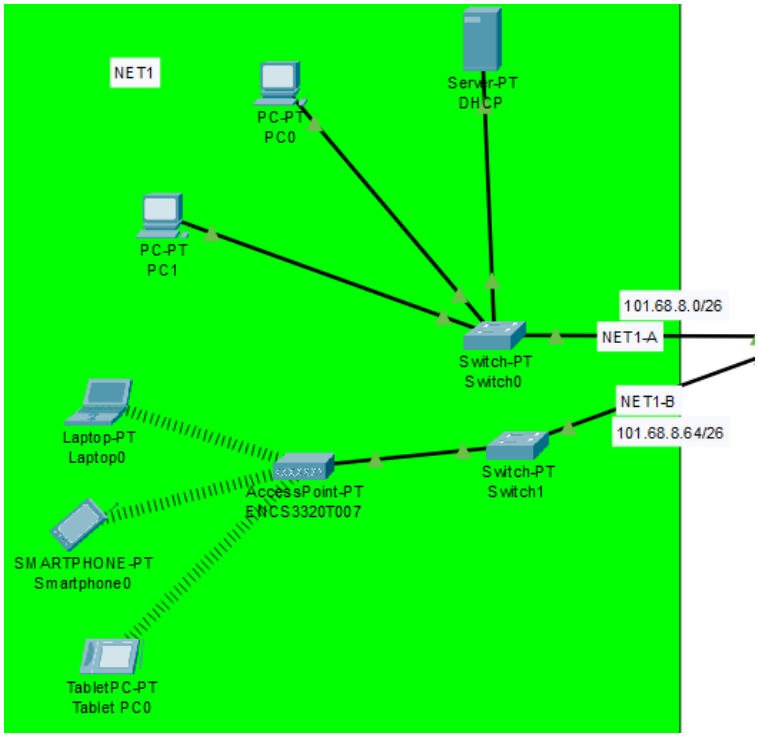


Figure 25: University network (Area 1).

1. **NET1-A:**

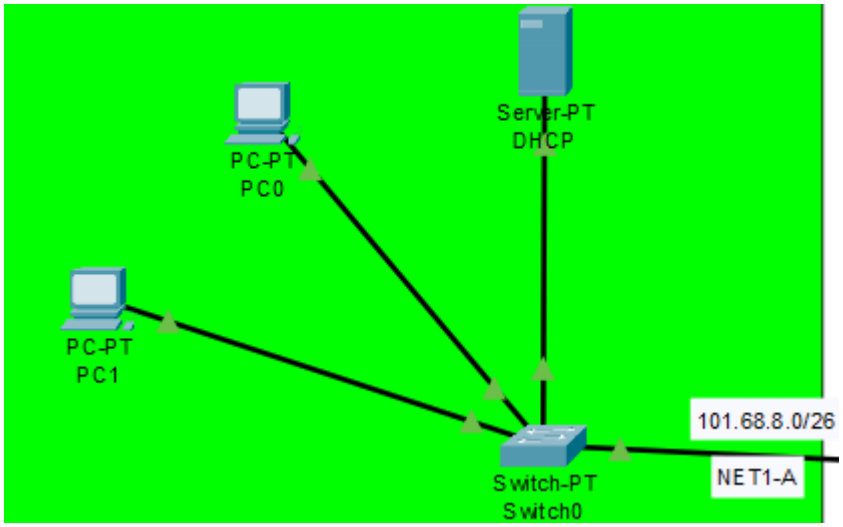


Figure 26: NET1-A.

- **<u>DHCP Server:</u>**

In this server, only the DHCP service is enabled. For this server we allocate a static IP address which is 101.68.8.2, as shown in the following figure:

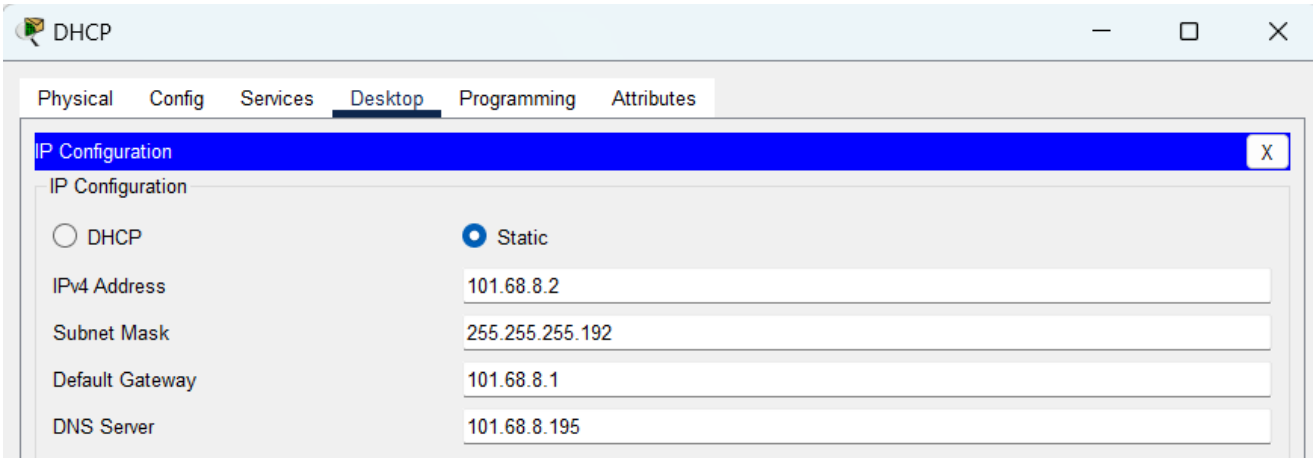


Figure 27: IP Configuration for DHCP Server.

Then, we created one pool for the NET1-A subnet (T007), as following:

T007:

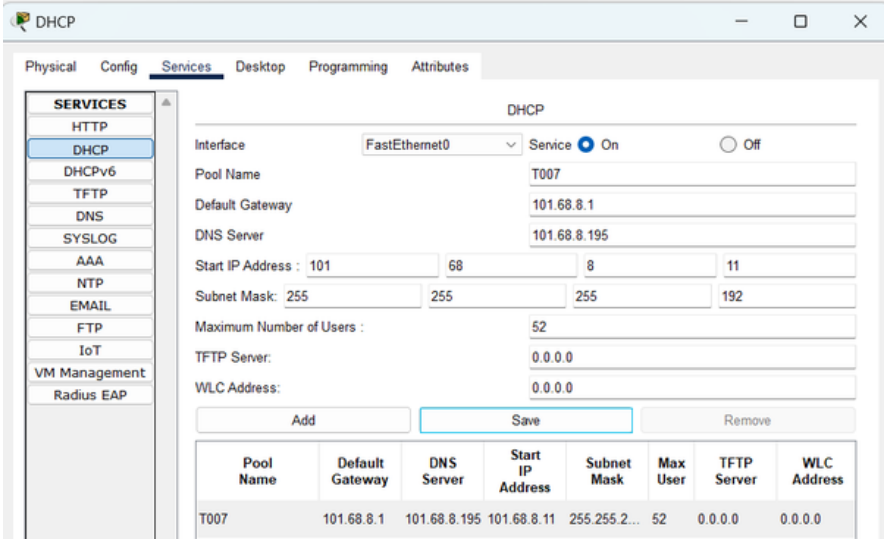


Figure 28: DHCP service with T007pool in DHCP Server.

Users is the total number of hosts that NET1_A network can include excluding 10 hosts, so the maximum number of users is (62 – 10) = 52 users.

PC0 and PC1 got their Ips dynamically from DHCP server.

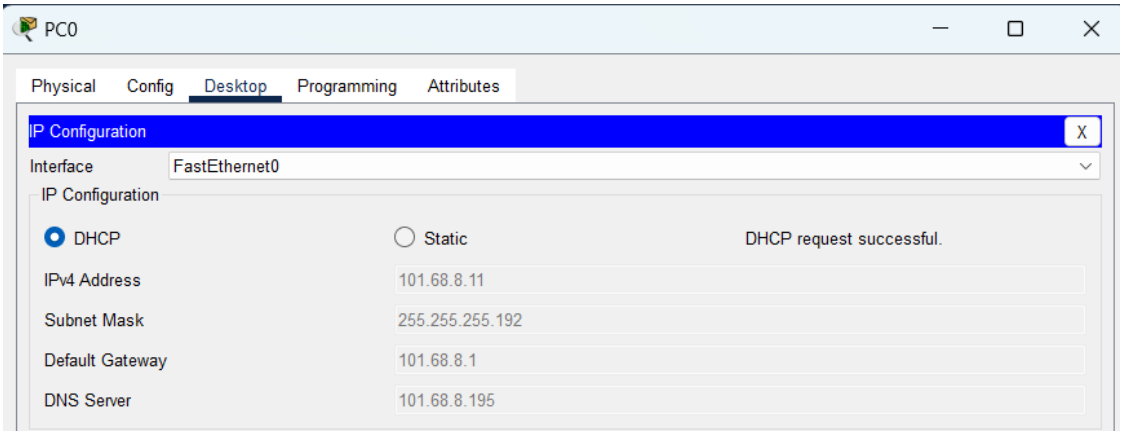


Figure 29:pc0 IP

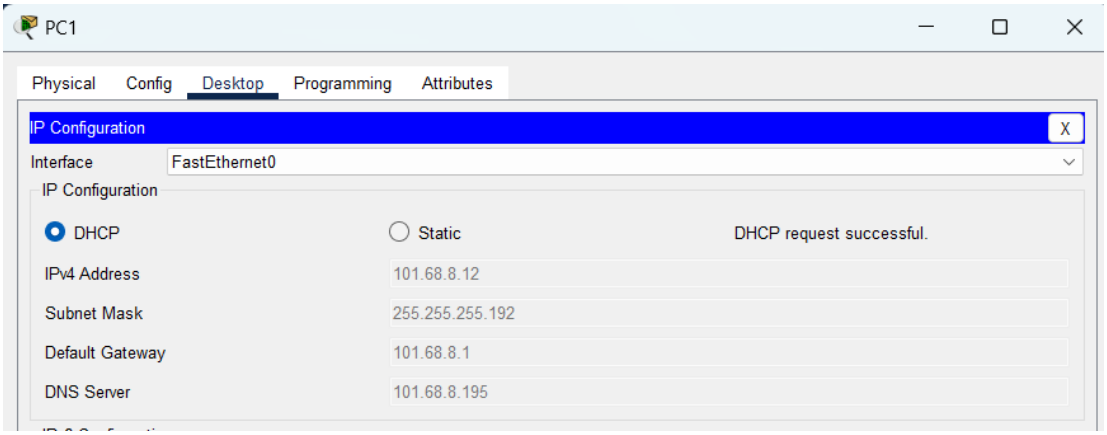


Figure 30:pc1 IP

2. **NET1-B:**

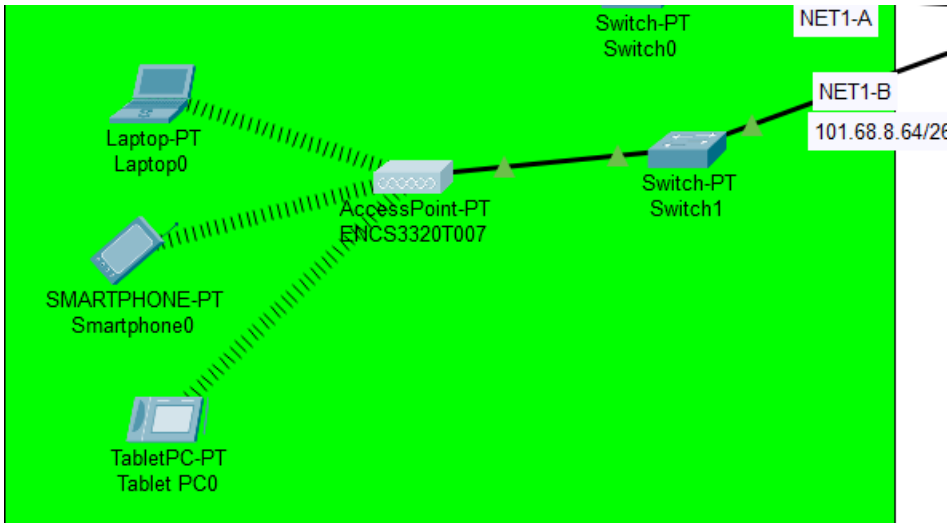


Figure 31: NET1-B.

Wireless Network Configuration:

- **SSID:** The network name is set to "ENCS3320T007", allowing users to identify and connect to the correct wireless network.

- **Security Model:** The network uses WPA2-PSK (Wi-Fi Protected Access 2 - Pre-Shared Key), offering strong security by requiring a password for access and encrypting the connection.

- **Encryption Type:** AES (Advanced Encryption Standard) is used to encrypt the data transmitted over the network, ensuring data privacy and protection against unauthorized access.

- **Network Key:** The password "Home_T007" is required to join the network, ensuring that only authorized users can connect.

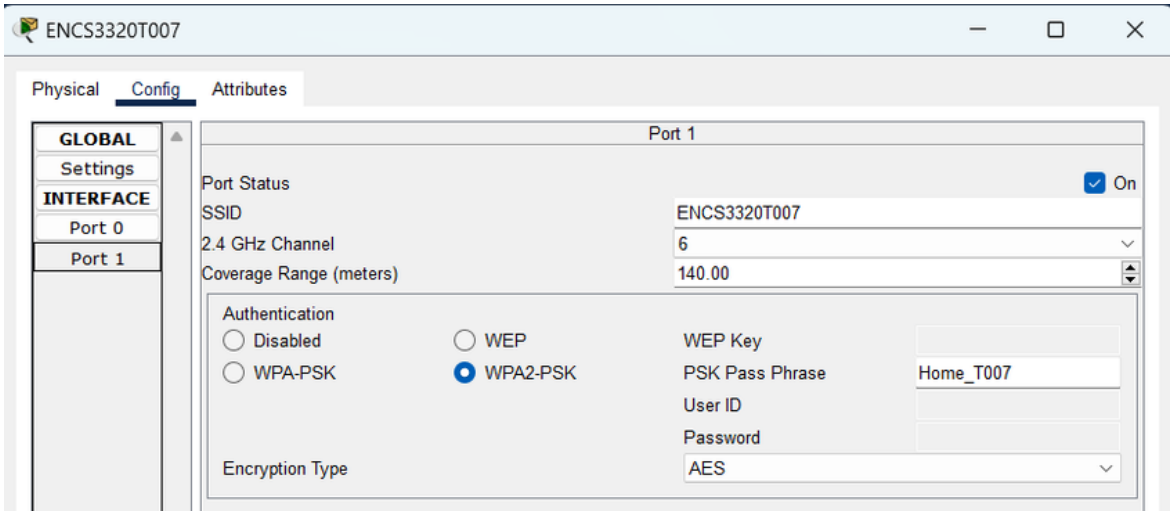


Figure 32: Wireless Network configuration.

Then we connect the end devices to the access point (wirelessly) and assigned static Ips for each one.

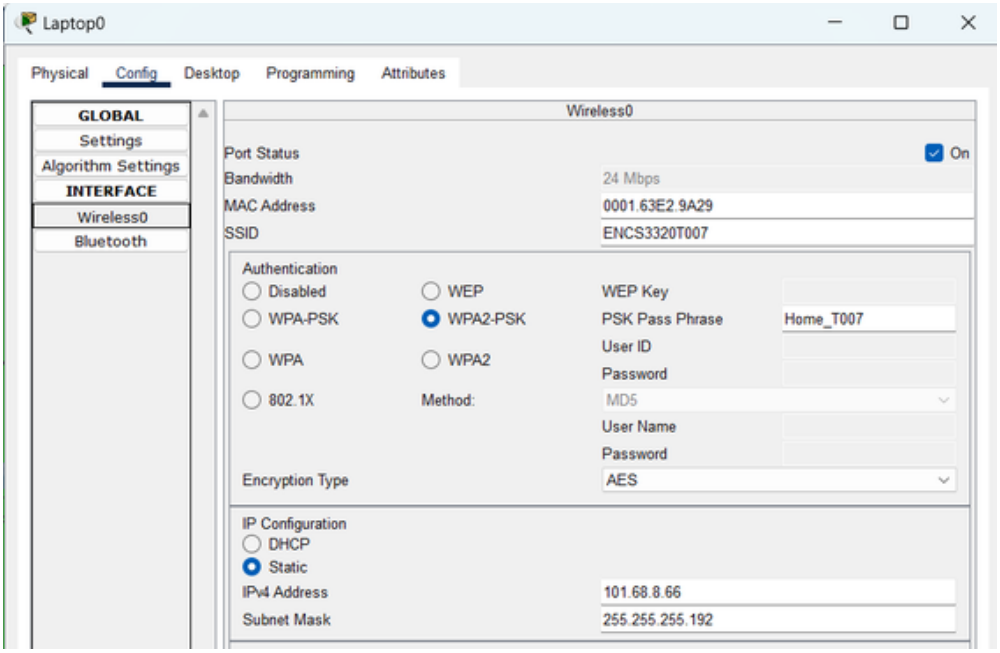


Figure 33: Laptop0 configuration.

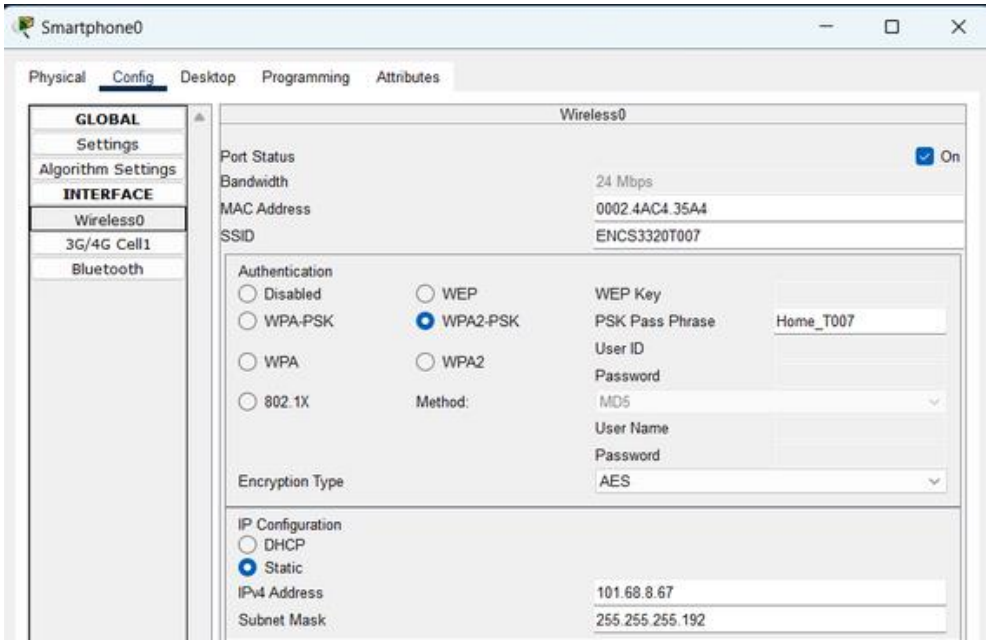


Figure 34: Smartphone0 configuration.

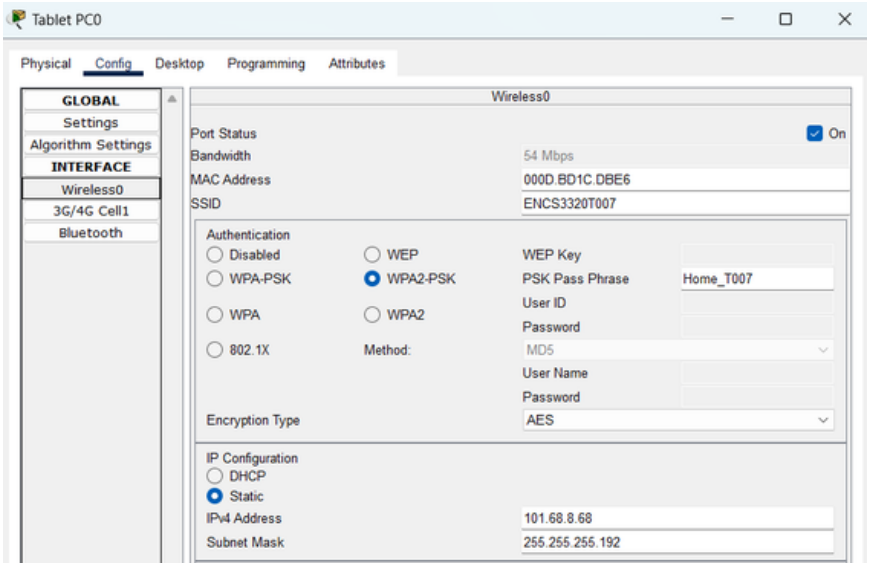


Figure 35: Tablet0 configuration.

We also configured each device with the default gateway and DNS server IP addresses.

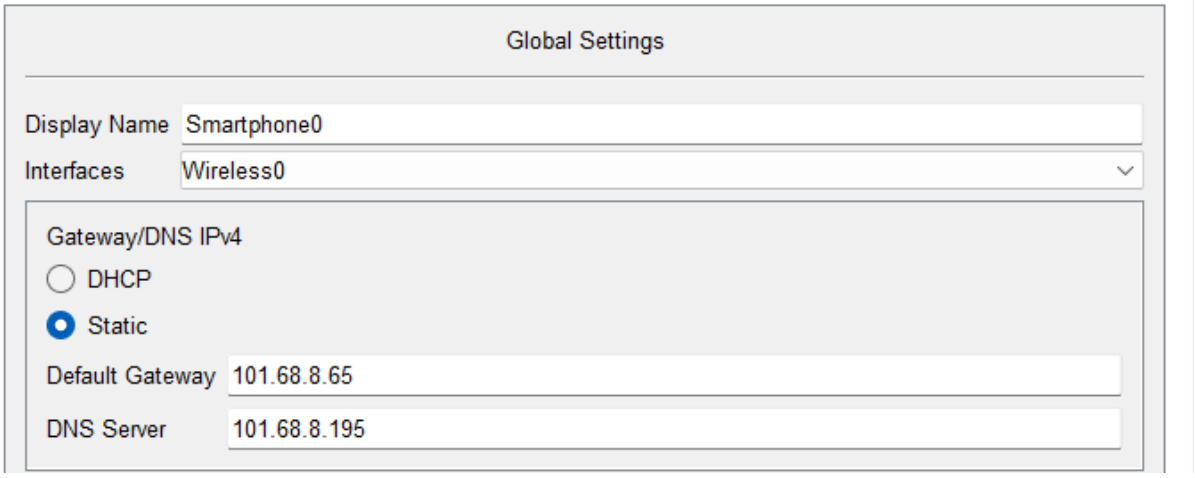


Figure 36: Smartphone0 setting.

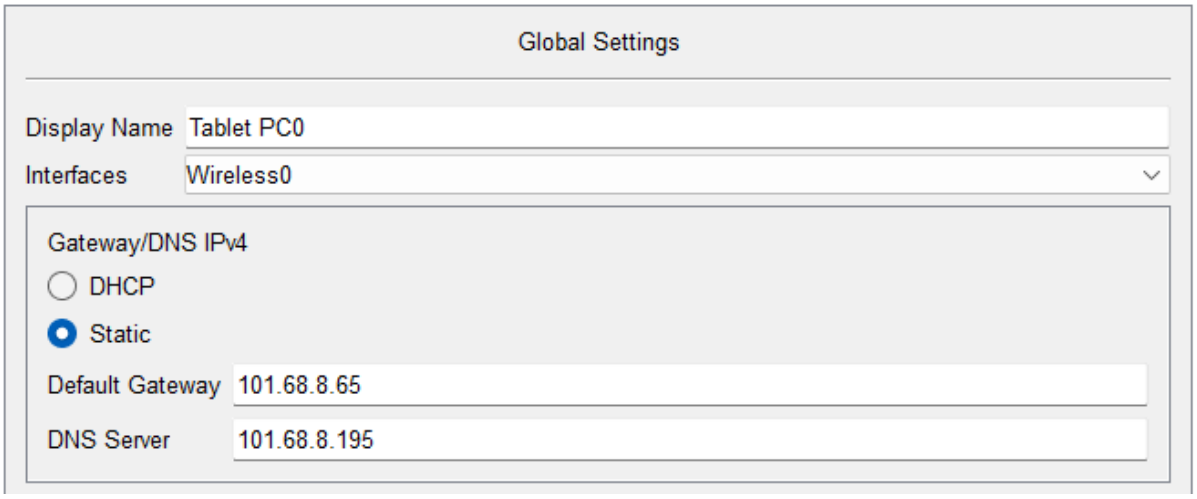


Figure 37: Tablet PC0 setting.

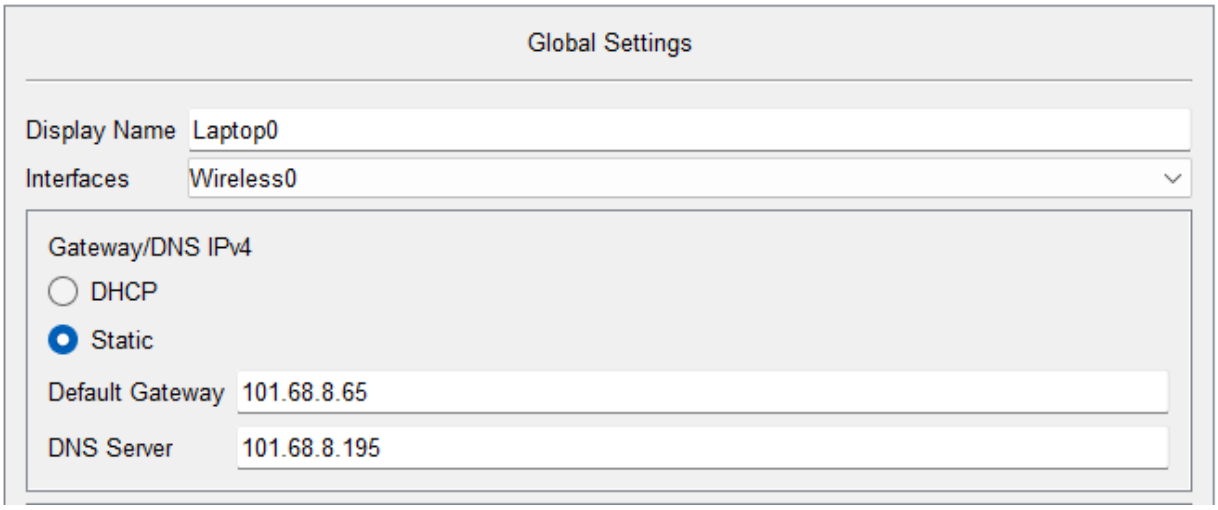


Figure 38: Laptop0 setting.

- **Street network (Area 2):**

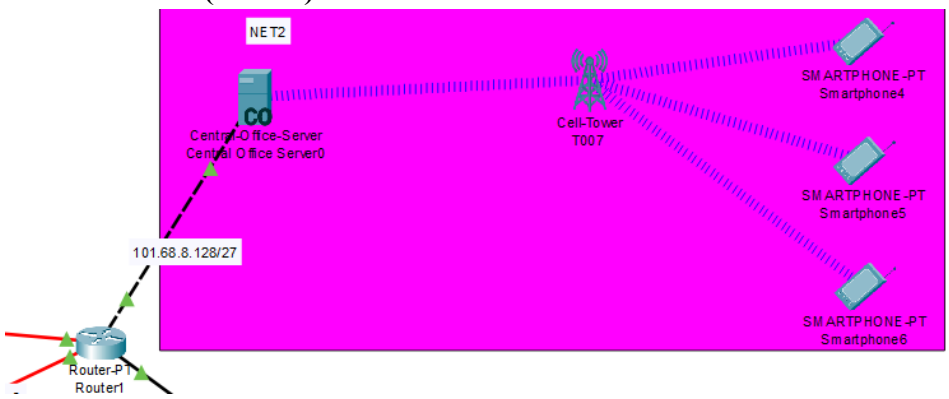


Figure 39: NET2.

- **Central-Office Server (CO Server):**

We set the provider's name T007:

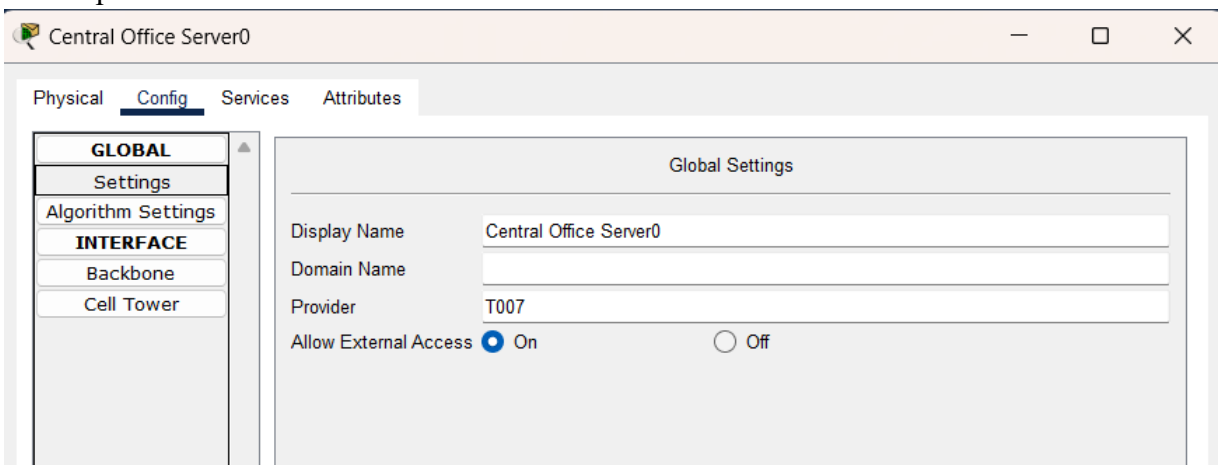


Figure 40: Provider name.

Then we set the IP to connect the CO server to the cell tower:

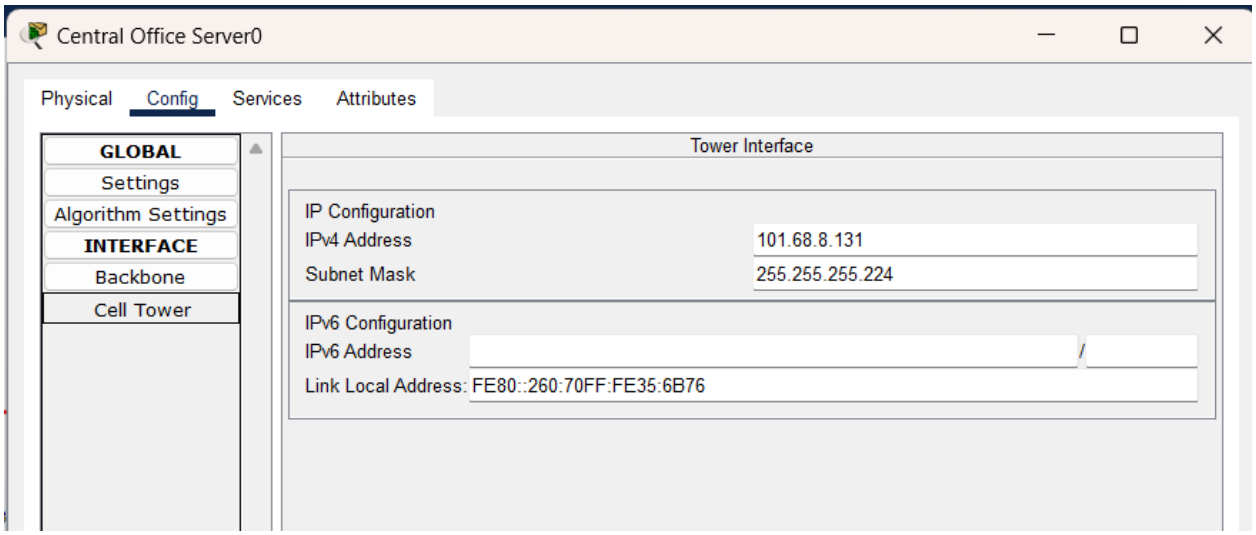


Figure 41: CO Server IP configuration.

Then we try to connect the CO server to the router via the backbone setting but it failed.

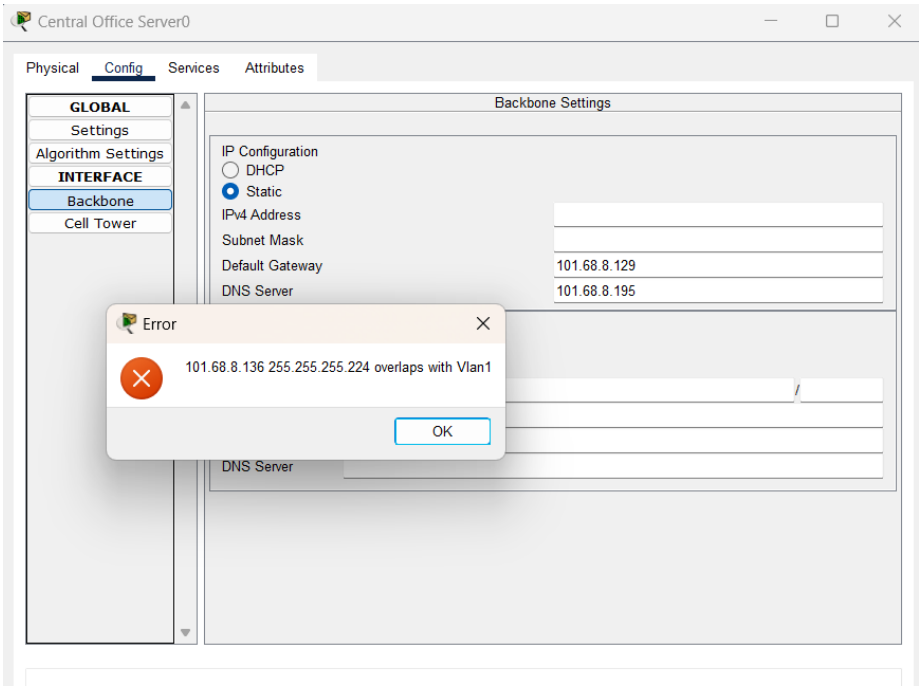


Figure 42: Failed result.

Smartphone4, 5, 6 got their Ips dynamically from the CO server.

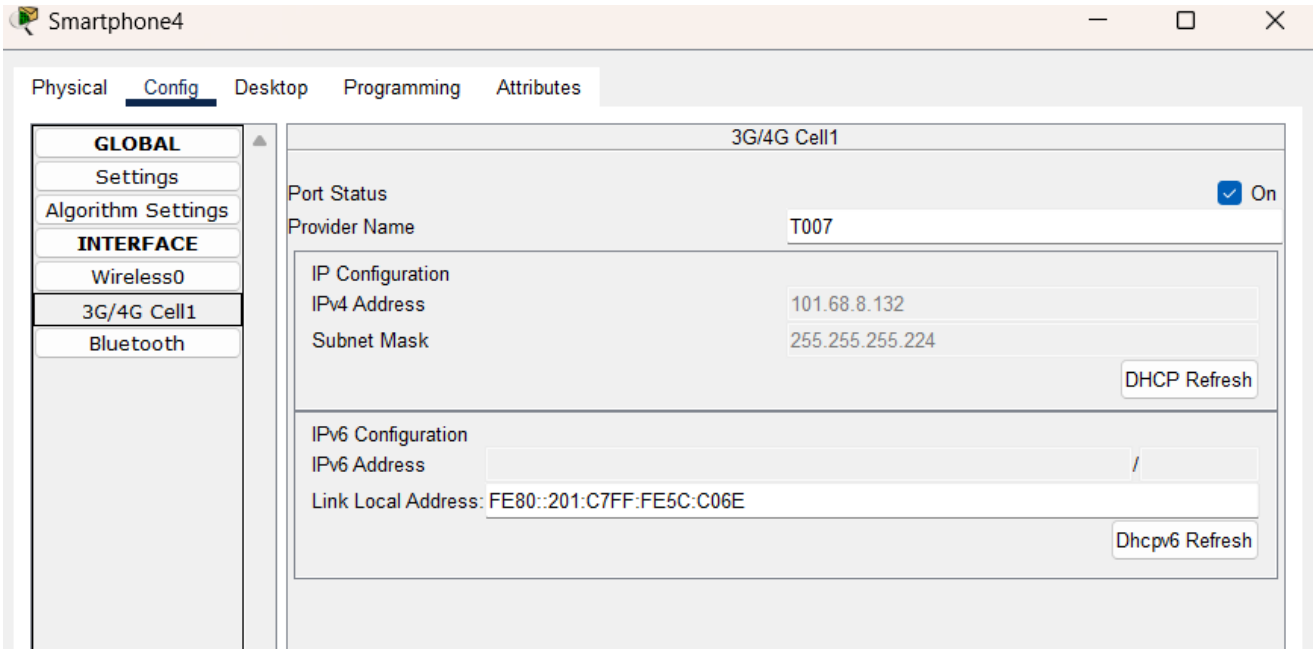


Figure 43: Smartphone4 IP.

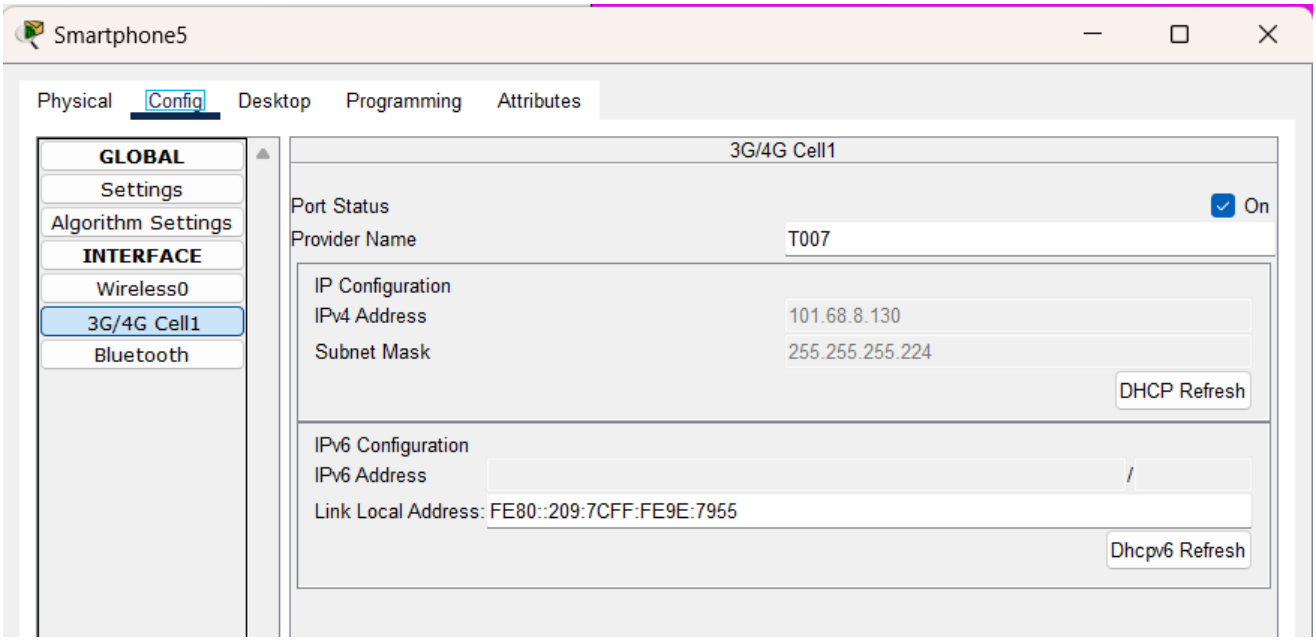


Figure 44: Smartphone5 IP.

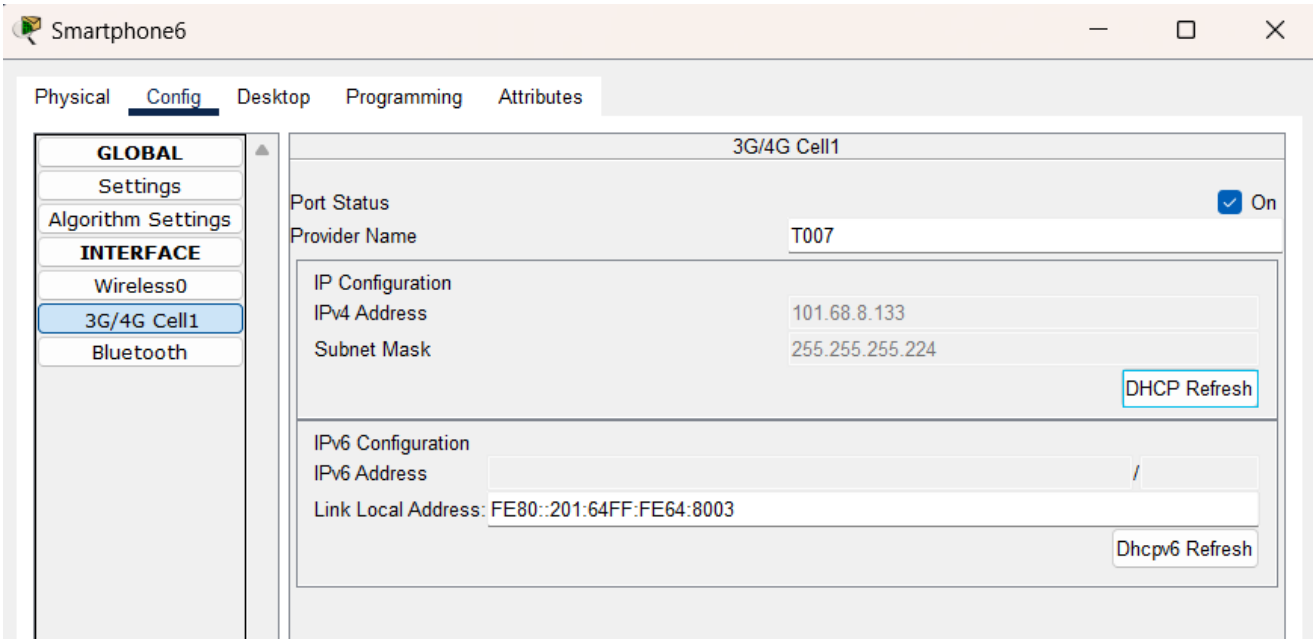


Figure 45: Smartphone6 IP.

- **Home network (Area 3):**

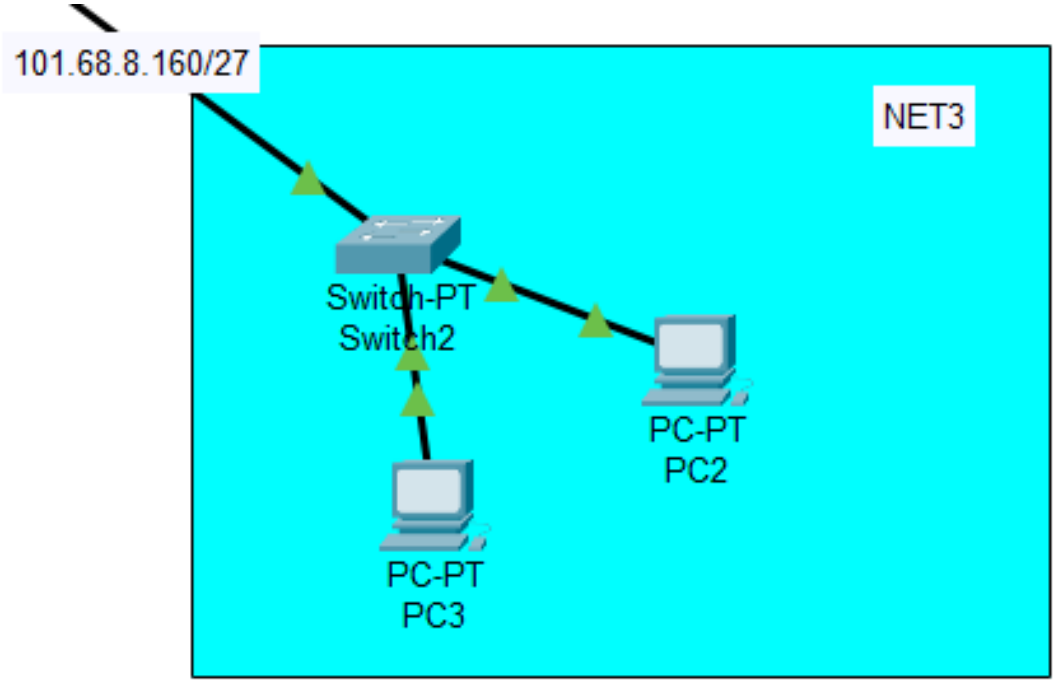


Figure 46: NET3.

In the figure above (NET3), a small wired network is shown with two PCs (PC2 and PC3) connected through a switch. Each device in this network is assigned a **static IP address** within the subnet 101.68.8.160/27.

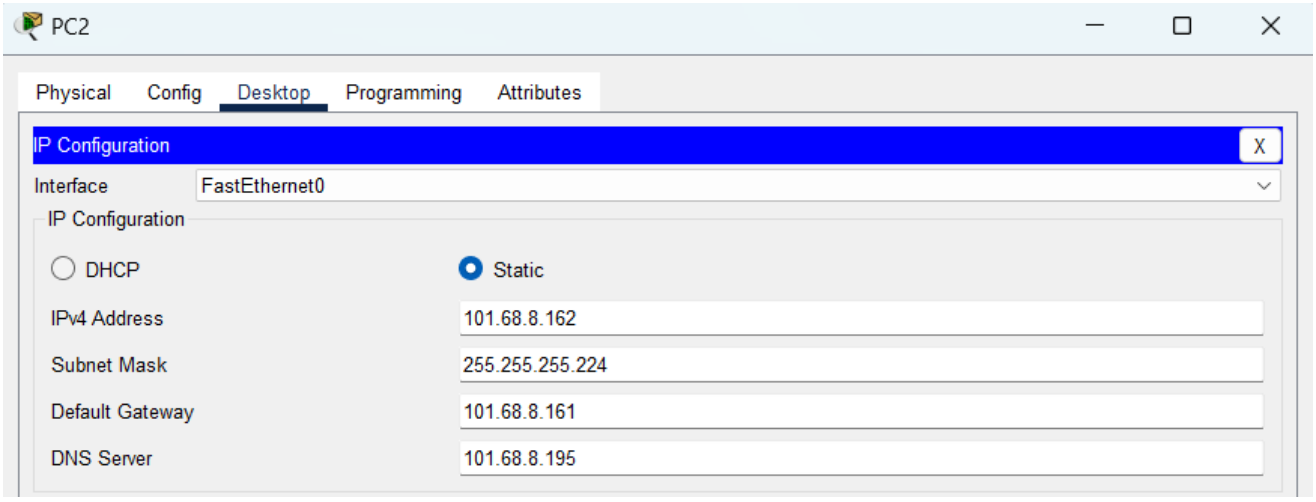


Figure 47: PC2 IP configuration.

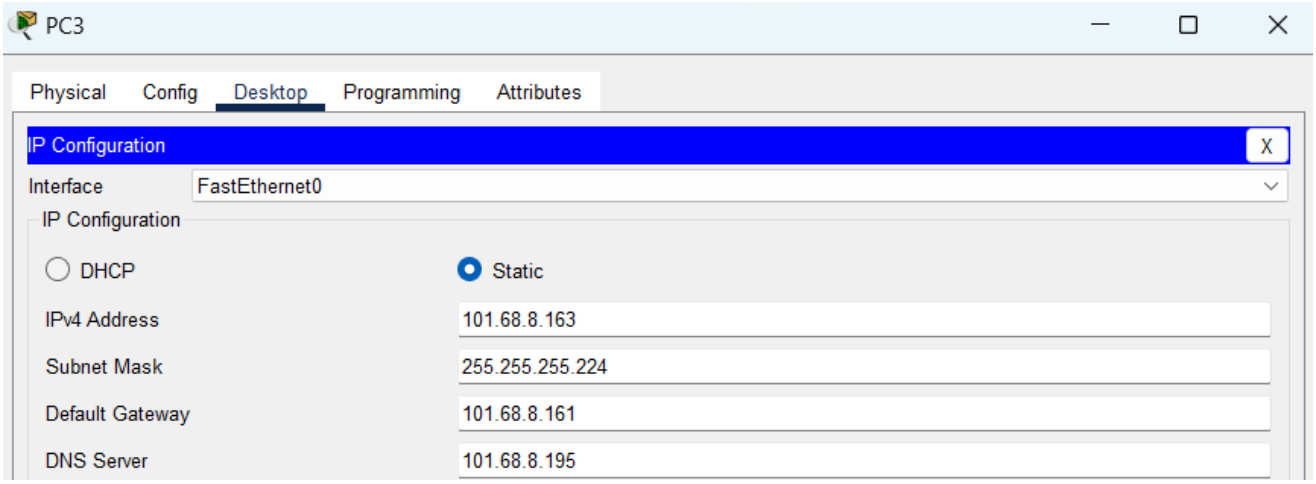


Figure 48: PC3 IP configuration.

- **Datacenter network (Area 4):**

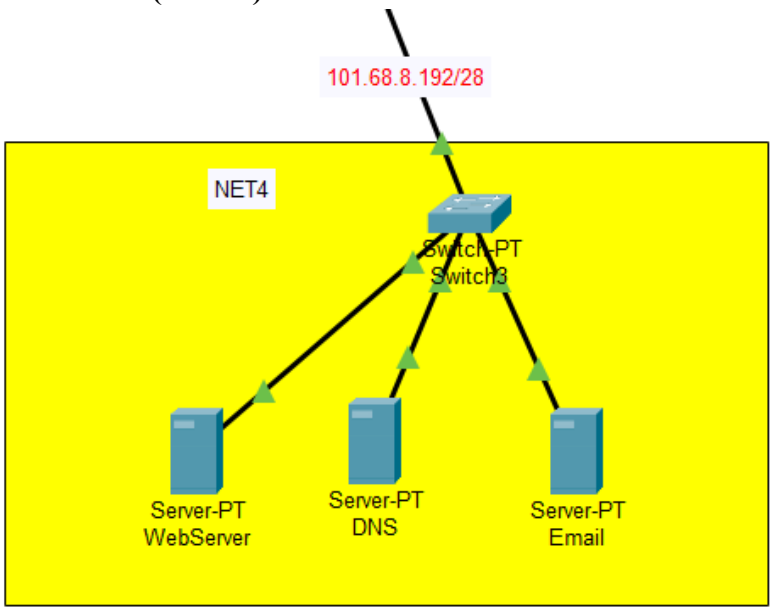


Figure 49: NET4.

In the figure above (NET4), a small wired server network is shown with three servers—Web Server, DNS Server, and Email Server—connected through a switch. Each server is assigned a static IP address within the subnet 101.68.8.192/28.

⭐ Servers Subnet

1. Web Server (www.coe.birzeit.edu):

In this server, only the HTTP and HTTP secure (HTTPS) protocols are enabled. For this server we allocate a static IP address which is 101.68.8.194, as shown in the following figure:

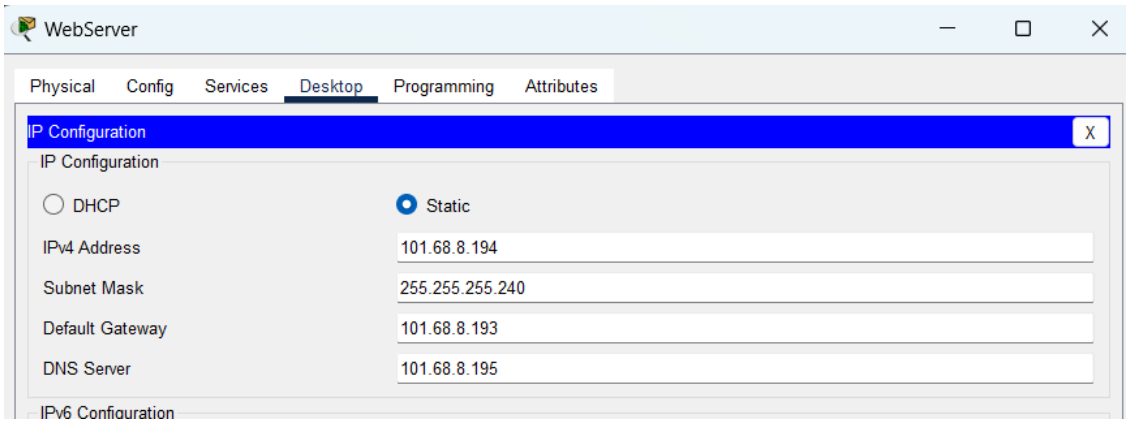


Figure 50: IP Configuration for Web Server.

In this server, HTTP and HTTPS Protocols are enabled, and we add an index.html page with 3 images (nagham.png, alzahraa.png, Ayshah.png) to this server, as shown in the following figure:

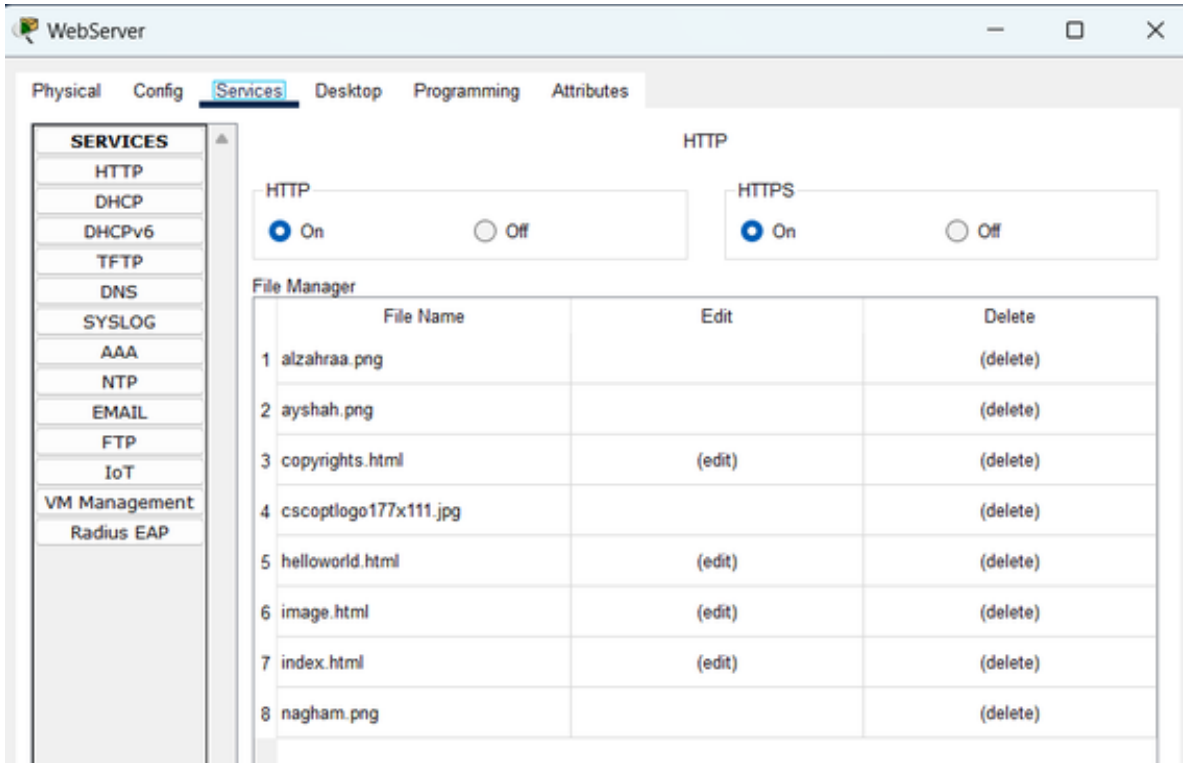
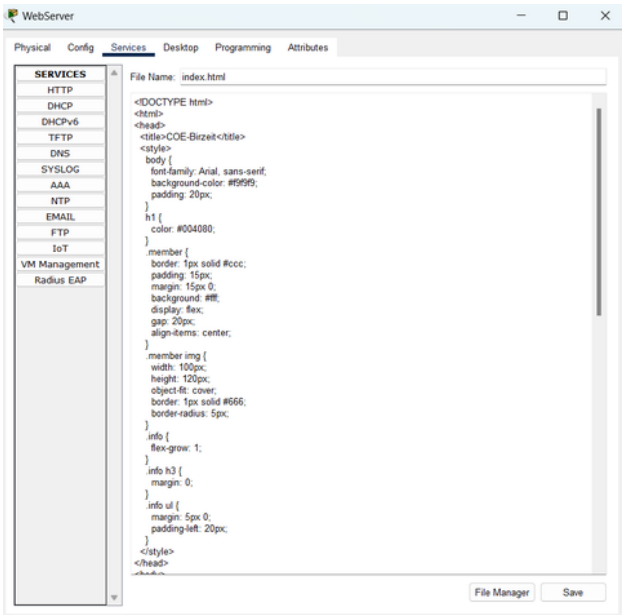


Figure 51:Services Configuration for Web Server.

Code added in index.html.



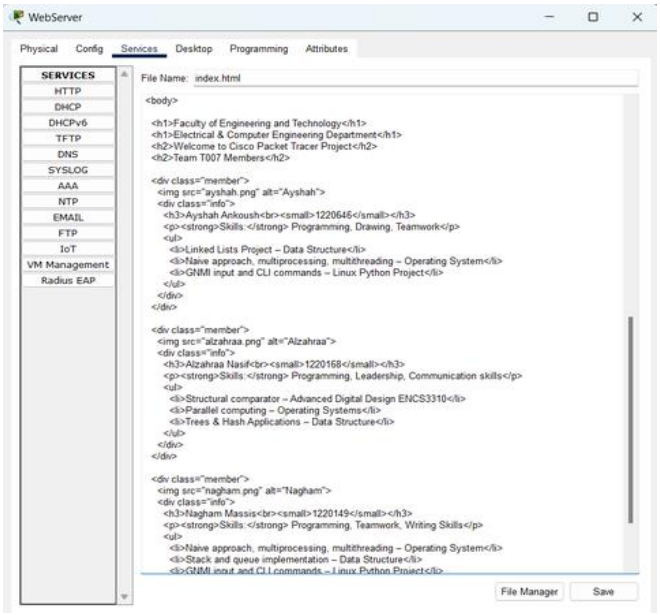


Figure 52: index.html code.

2. Email Server (mail.coe.birzeit.edu):

In this server, only the SMTP (for sending emails) and POP3 (for receiving emails) services are enabled. For this server we allocate a static IP address which is 101.68.8.196, and set the domain name (mail server alias hostname) as coe.birzeit.edu.

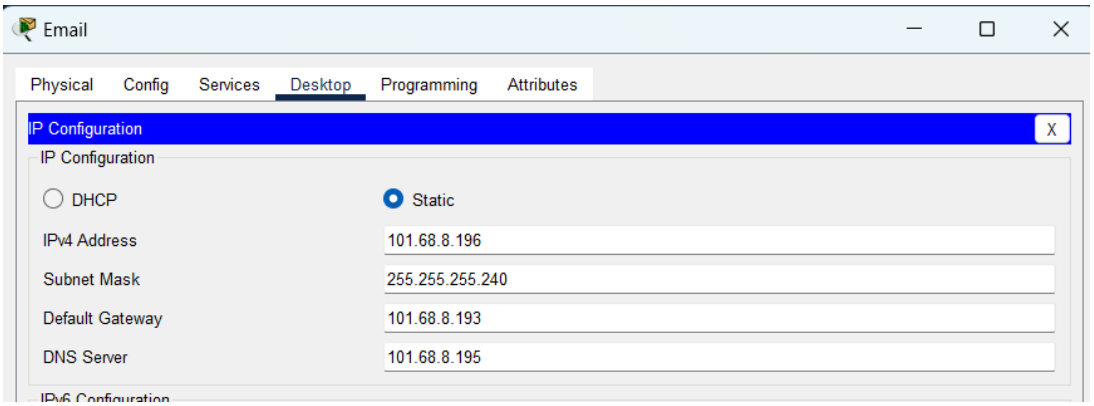


Figure 53: IP configuration for Email Server.

Then, we create a user account for each member of our team with the following format:

Username {HomeSID, StreetSID, UniversitySID}.	Password {Team member's SID}
Home1220149	1220149
Street1220646	1220646
University1220168	1220168

Table 3:user accounts for Email server

As shown in the figure below:

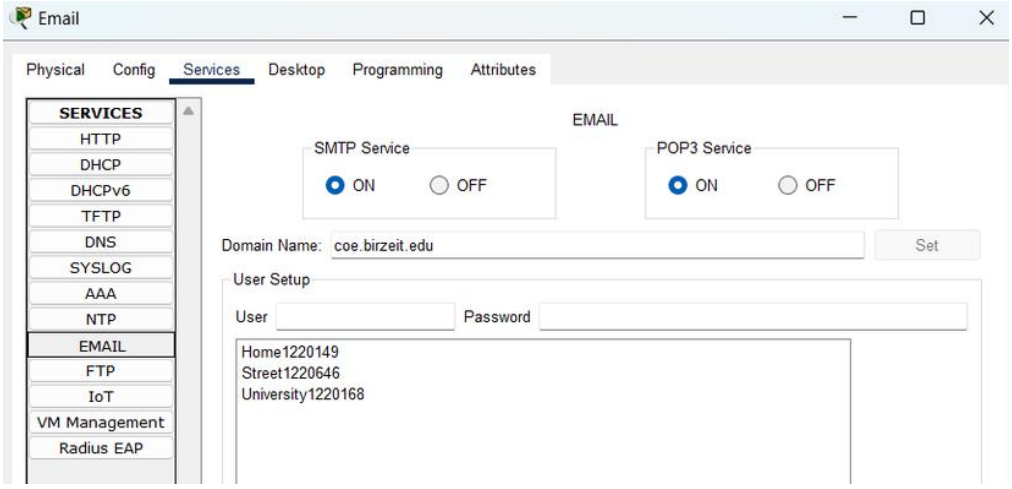


Figure 54: Email service with the user setup in Email Server.

3. DNS Server (dns.coe.birzeit.edu):

In this server, only the DNS service is enabled. For this server we allocate a static IP address which is 101.68.8.195, as shown in the following figure:

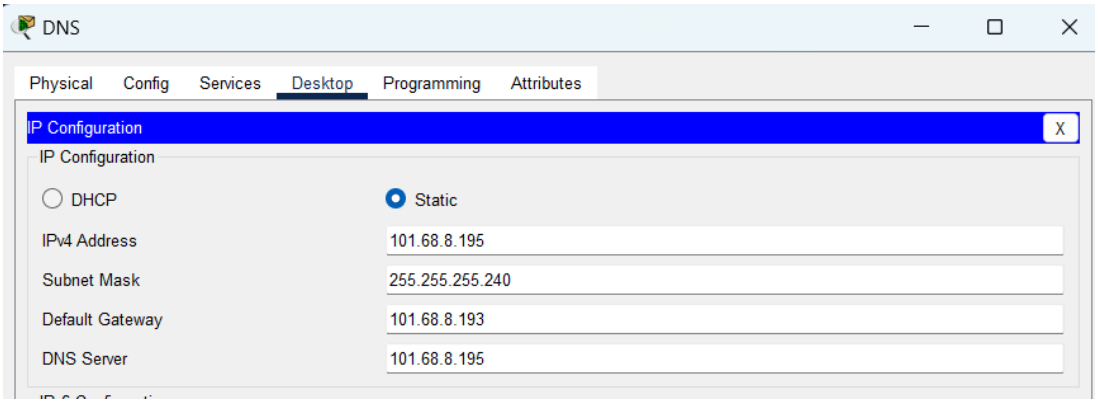


Figure 55: IP Configuration for DNS Server.

Then, we added the following Resource Records (RRs):

Name	Value	Type
www.coe.birzeit.edu	The IP address of the *www.coe.birzeit.edu* server	A
mail.coe.birzeit.edu	The IP address of the *mail.coe.birzeit.edu* server	A
coe.birzeit.edu	mail.coe.birzeit.edu	CNAME

Table 4: Resource Records (RRs).

As shown in the figure below:

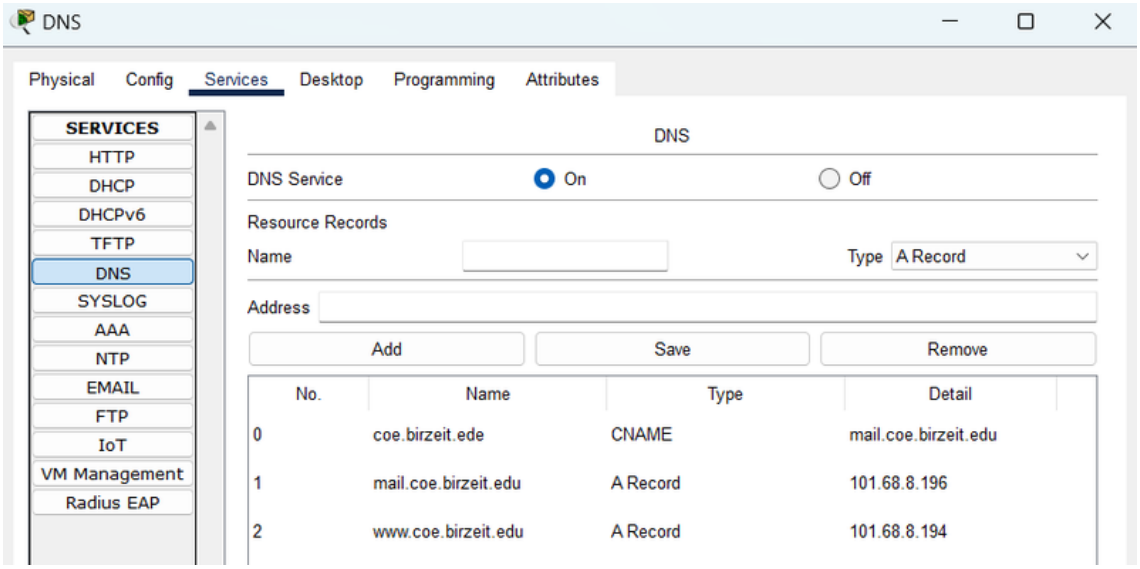


Figure 56: DNS service with the Records (RRs) in the DNS Server.

These RRs map the Web server www.coe.birzeit.edu, the email server mail.coe.birzeit.edu, and the alias domain coe.birzeit.edu to their respective IP addresses or canonical names. Additionally, they set the hostname of the authoritative name server for the domain coe.birzeit.edu to mail.coe.birzeit.edu and set the DNS server to resolve the alias hostname coe.birzeit.edu to its corresponding canonical name mail.coe.birzeit.edu. This configuration ensures proper domain name resolution for core services such as email and web access within the network.

3. Testing

❖ **Access to www.coe.birzeit.edu from a Faculty of Engineering and Technology end device:**

1. **From PC0 in NET1-A:**

Figure 57: Successful access to www.coe.birzeit.edu from PC0.

2. **From smartphone in NET1-B:**

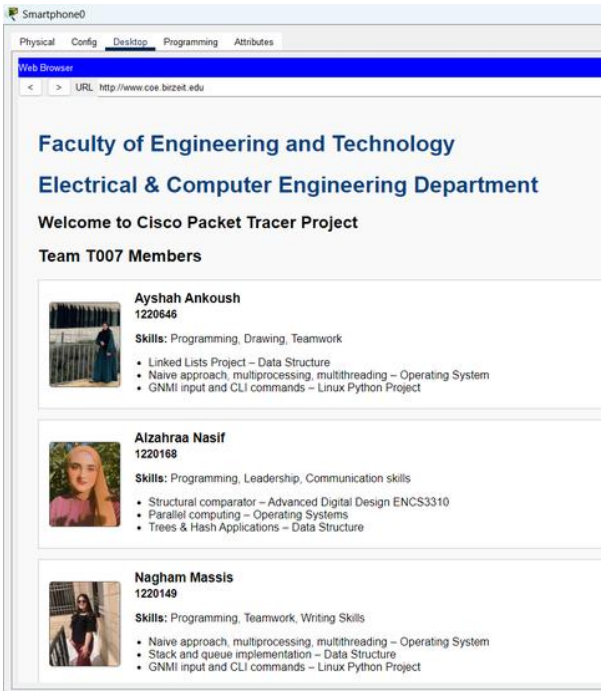


Figure 58: Successful access to www.coe.birzeit.edu from smartphone0.

3. From PC2 in NET3:

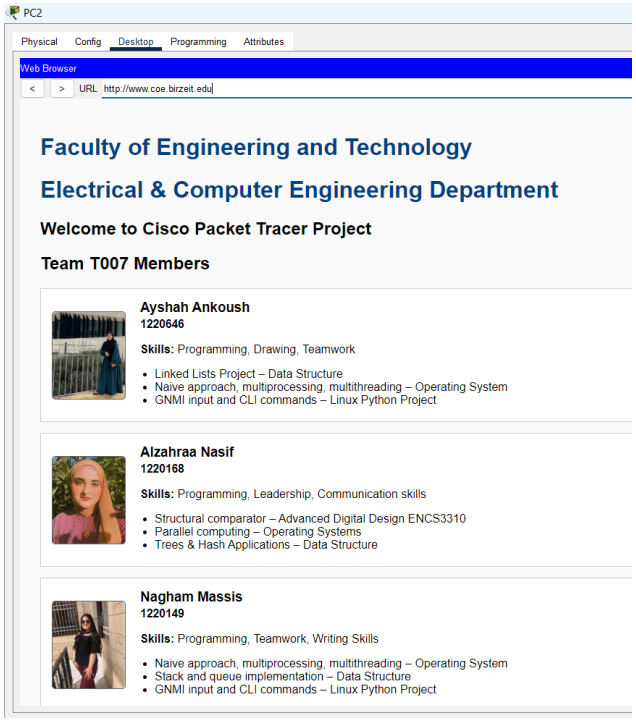


Figure 59: Successful access to www.coe.birzeit.edu from PC2.

❖ Sending and Receiving Emails:

1. From University to Home:

we configured PC3 email client in NET3 for the Home account, as shown in the following figure:

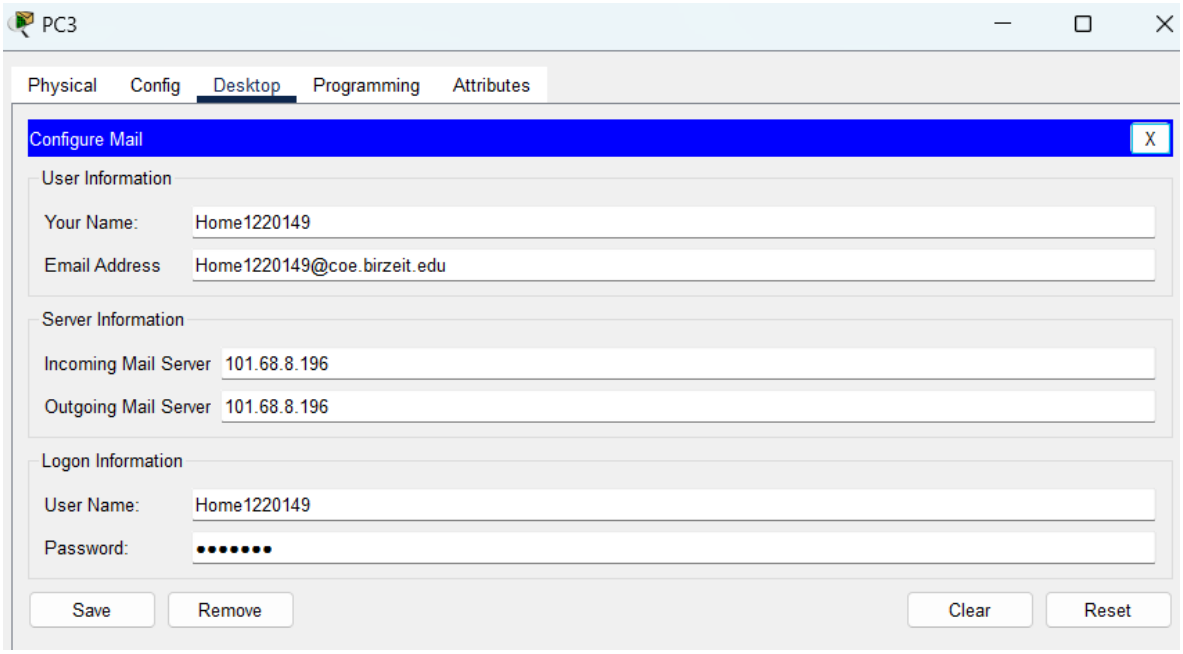


Figure 60: Email client Configuration for Home account (on PC3).

we configured PC1 email client in NET1-A for the University account, as shown in the following figure:

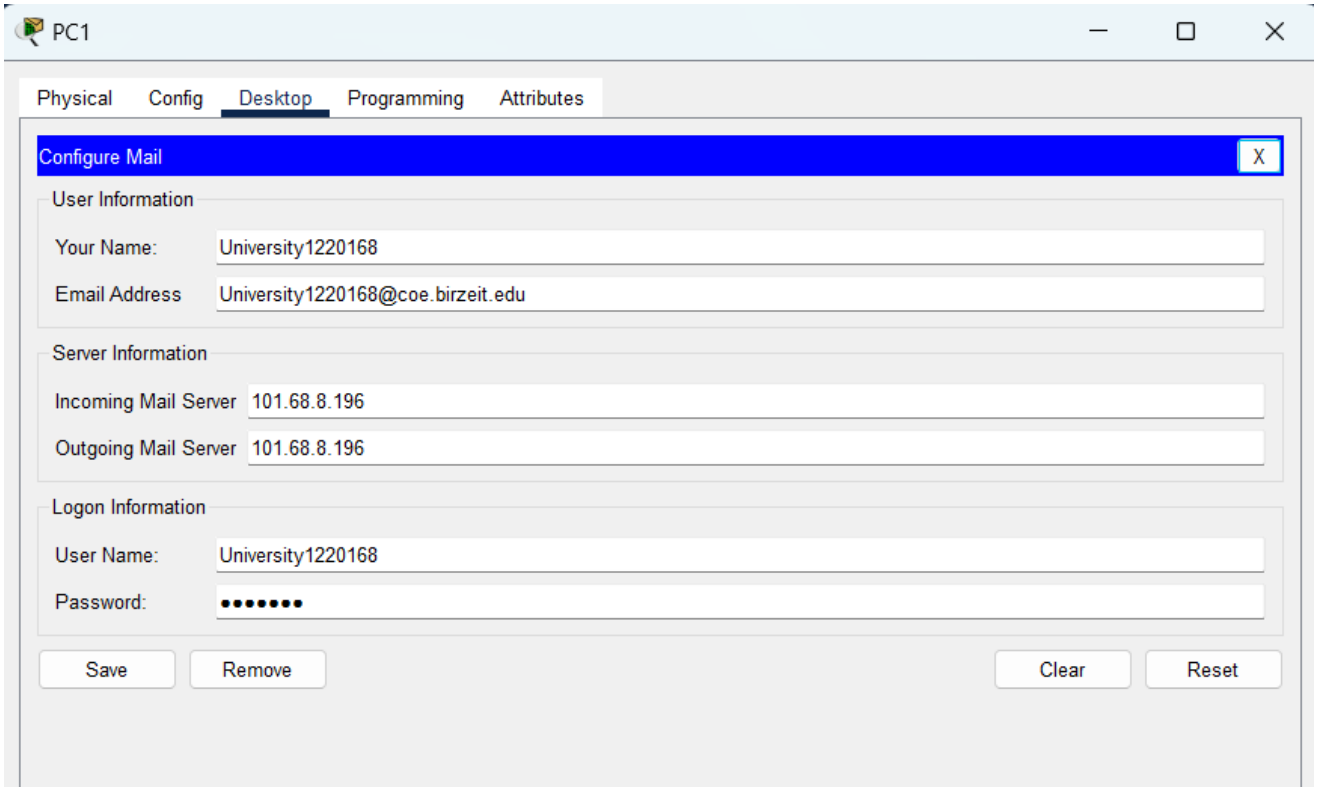


Figure 61: Email client Configuration for University account (on PC1).

Then we tried to send email from University account to Home account:

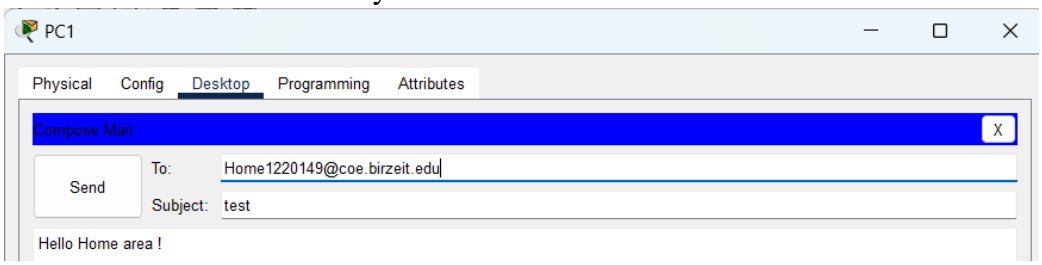


Figure 62: Email to be sent from email account to another email account.

The status of sending this Email is success, as shown in the following figure:

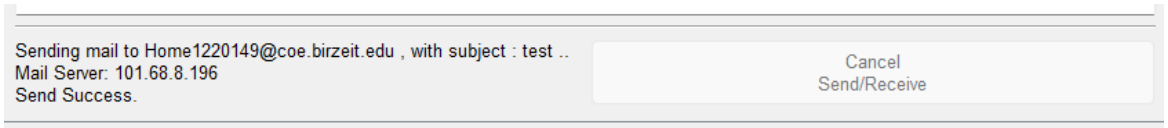


Figure 63: Status of sending email from email account to another email account.

The status of Receiving the Email in the other account (Home1220149@coe.birzeit.edu) which configured on PC1:

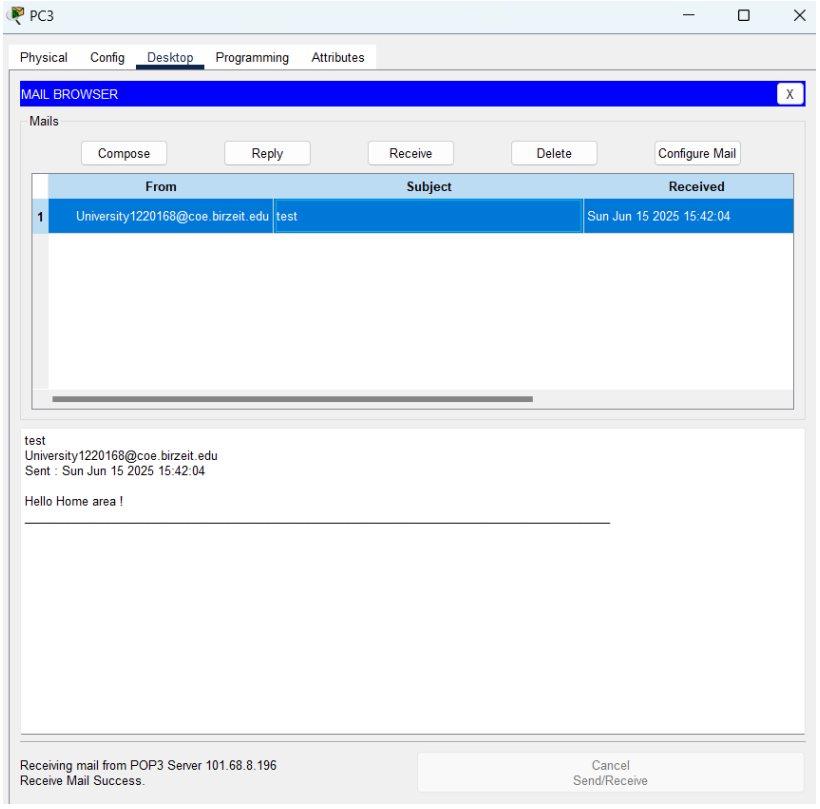


Figure 64: Receiving Email which sent from email account to another email account.

2. From Home to University:

we configured PC2 email client in NET3 for the Home account, as shown in the following figure:

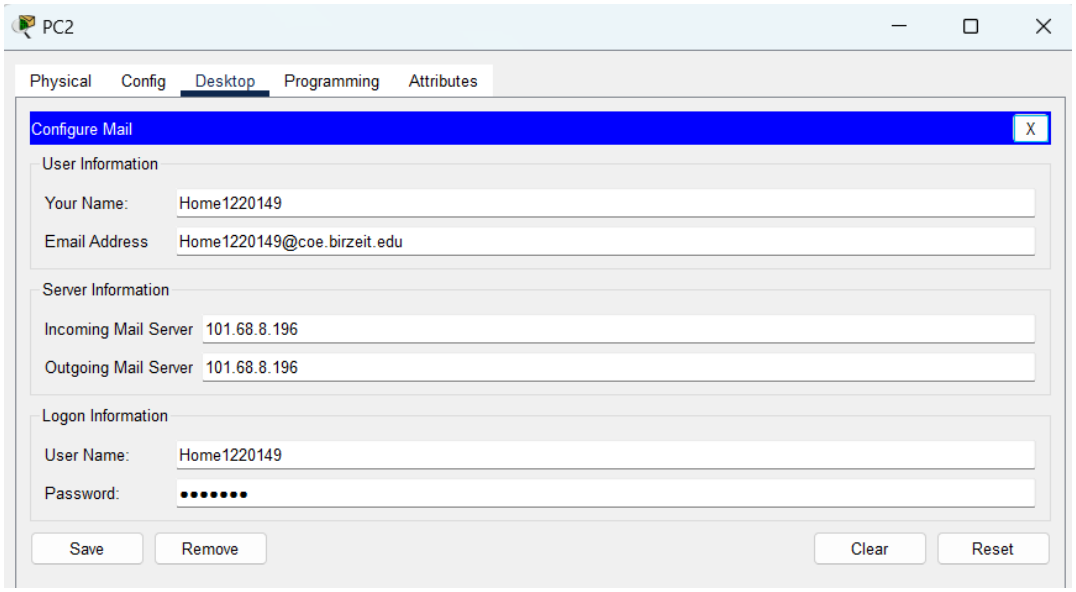


Figure 65: Email client Configuration for Home account (on PC2).

we configured Smartphone0 email client in NET1-B for the University account, as shown in the following figure:

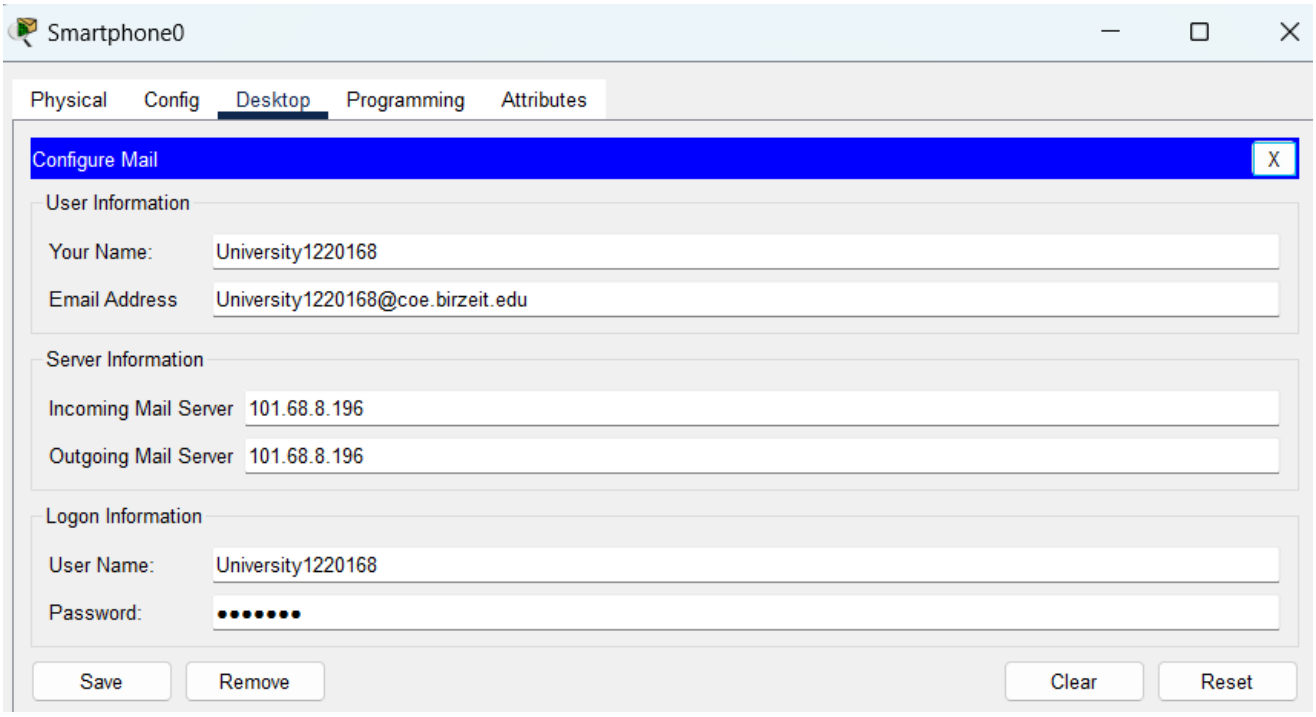


Figure 66: Email client Configuration for University account (on Smartphone0).

Then we tried to send email from Home account to University account:

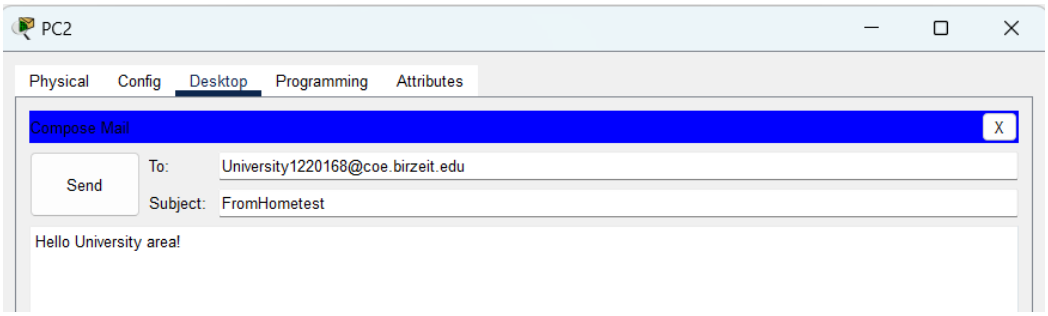


Figure 67: Email to be sent from email account to another email account.

Status of sending email from email account to another email account.

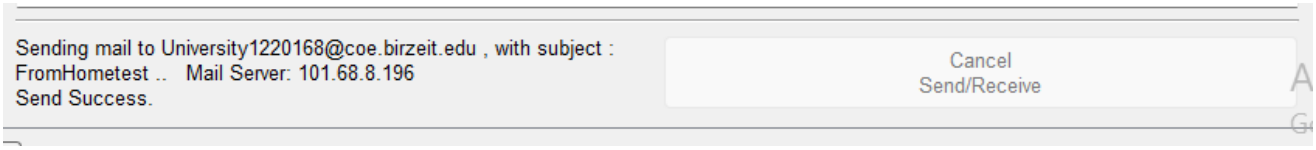


Figure 68: Status of sending email from email account to another email account.

The status of Receiving the Email in the other account (University1220168@coe.birzeit.edu) which configured on Smartphone0:

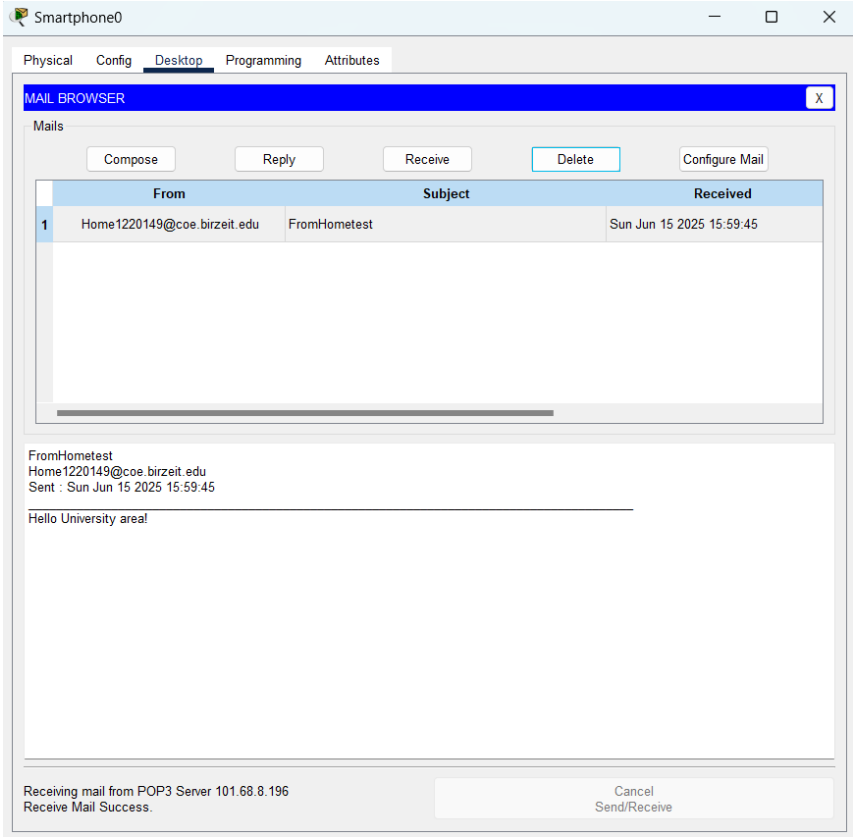


Figure 69: Receiving Email which sent from email account to another email account.

3. From Street to Home:

we configured Smartphone7 email client in NET2 for the Street account, as shown in the following figure:

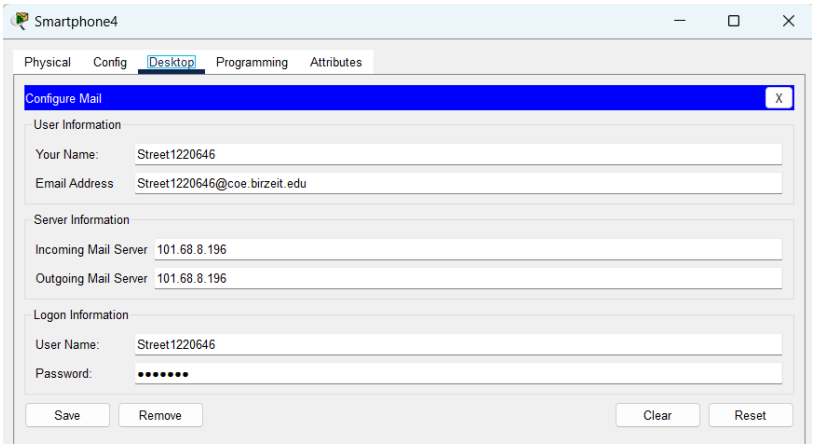


Figure 70: Email client Configuration for Street account (on Smartphone4).

Then we tried to send email from Street account to Home account:

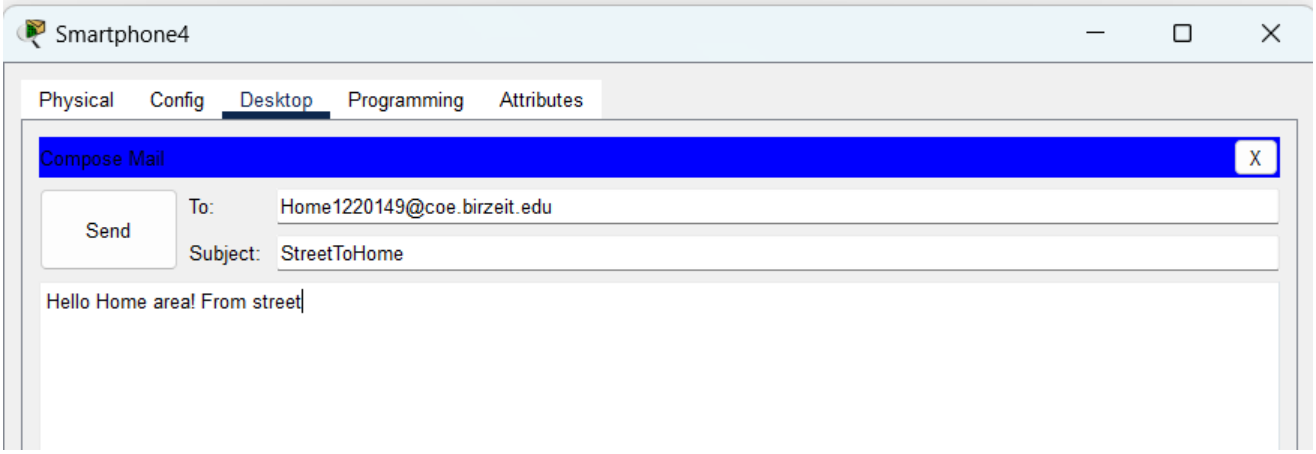


Figure 71: Email to be sent from email account to another email account.

Status of sending email from email account to another email account.

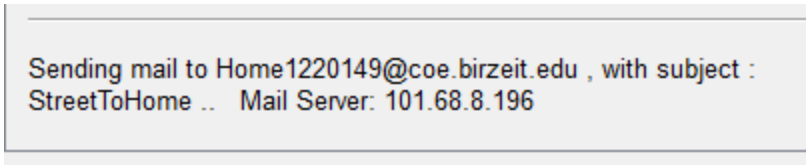


Figure 72: Status of sending email from email account to another email account.

The status of Receiving the Email in the other account (Street1220646@coe.birzeit.edu) which configured on Smartphone4 (Failed):

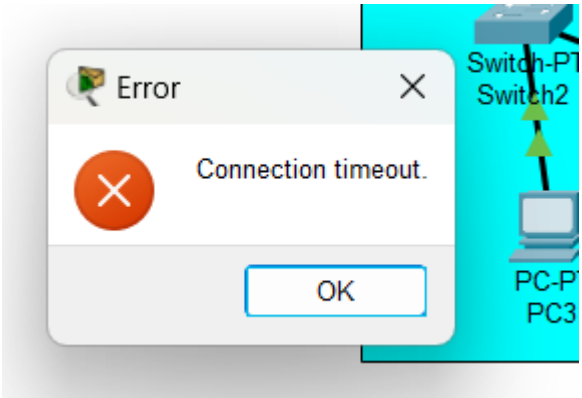


Figure 73: Failed receiving Email which sent from email account to another email account.

3. **<u>Pinging:</u>**

A. ping from pc0 to pc1:

```
C:\>ping 101.68.8.11

Pinging 101.68.8.11 with 32 bytes of data:

Reply from 101.68.8.11: bytes=32 time<1ms TTL=128
Reply from 101.68.8.11: bytes=32 time=9ms TTL=128
Reply from 101.68.8.11: bytes=32 time=9ms TTL=128
Reply from 101.68.8.11: bytes=32 time<1ms TTL=128

Ping statistics for 101.68.8.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 4ms
```

Figure 74: ping from pc0 to pc1.

B. ping from pc0 to laptop in NET1-B:

```
C:\>ping 101.68.8.66

Pinging 101.68.8.66 with 32 bytes of data:

Reply from 101.68.8.66: bytes=32 time=18ms TTL=127
Reply from 101.68.8.66: bytes=32 time=3ms TTL=127
Reply from 101.68.8.66: bytes=32 time=13ms TTL=127
Reply from 101.68.8.66: bytes=32 time=7ms TTL=127

Ping statistics for 101.68.8.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 18ms, Average = 10ms
```

Figure 75: ping from pc0 to laptop in NET1-B.

C. ping from PC0 to smartphone in NET1-B:

```
C:\>ping 101.68.8.67

Pinging 101.68.8.67 with 32 bytes of data:

Reply from 101.68.8.67: bytes=32 time=20ms TTL=127
Reply from 101.68.8.67: bytes=32 time=7ms TTL=127
Reply from 101.68.8.67: bytes=32 time=10ms TTL=127
Reply from 101.68.8.67: bytes=32 time=13ms TTL=127

Ping statistics for 101.68.8.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 20ms, Average = 12ms
```

Figure 76: ping from PC0 to smartphone in NET1-B.

D. ping from PC0 to PC1 in NET3:

```
C:\>ping 101.68.8.162

Pinging 101.68.8.162 with 32 bytes of data:

Reply from 101.68.8.162: bytes=32 time=11ms TTL=126
Reply from 101.68.8.162: bytes=32 time=9ms TTL=126
Reply from 101.68.8.162: bytes=32 time=13ms TTL=126
Reply from 101.68.8.162: bytes=32 time=8ms TTL=126

Ping statistics for 101.68.8.162:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 13ms, Average = 10ms
```

Figure 77: ping from PC0 to PC1 in NET3.

E. ping from PC0 to DNS server in NET4:

```
C:\>ping 101.68.8.195

Pinging 101.68.8.195 with 32 bytes of data:

Reply from 101.68.8.195: bytes=32 time=14ms TTL=126
Reply from 101.68.8.195: bytes=32 time=11ms TTL=126
Reply from 101.68.8.195: bytes=32 time=10ms TTL=126
Reply from 101.68.8.195: bytes=32 time=12ms TTL=126

Ping statistics for 101.68.8.195:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 14ms, Average = 11ms
```

Figure 78: ping from PC0 to DNS server in NET4.

F. ping from PC0 to smartphone4 in NET2 (faild):

```
C:\>ping 101.68.8.132

Pinging 101.68.8.132 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 101.68.8.132:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 79: ping from PC0 to smartphone4 in NET2 (faild).

G. ping from PC2 in NET3 to PC0 in NET1-A:

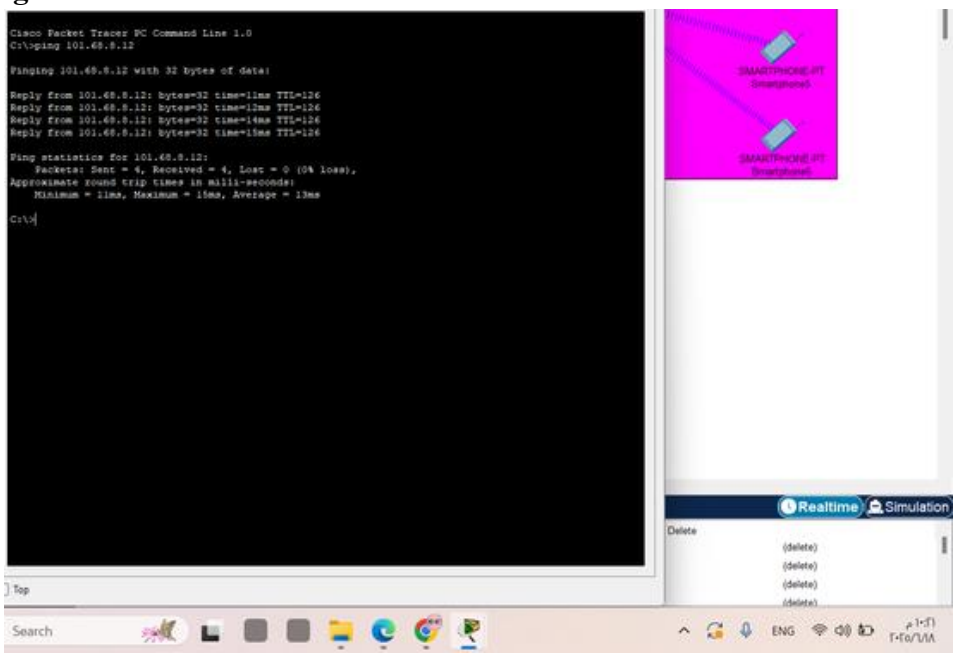
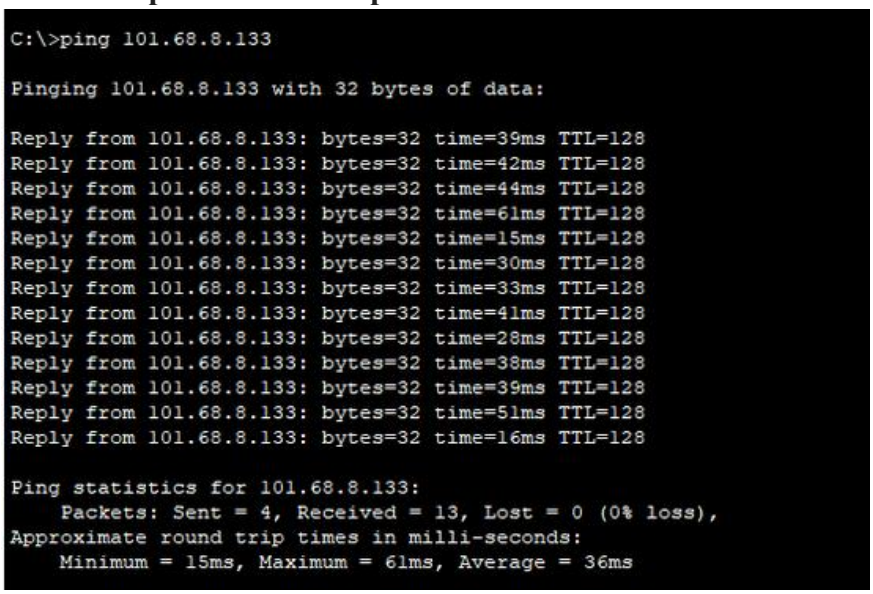


Figure 80: ping from PC2 in NET3 to PC0 in NET1-A.

H. ping from smartphone4 to smartphone6 in NET2:



```
C:\>ping 101.68.8.133

Pinging 101.68.8.133 with 32 bytes of data:

Reply from 101.68.8.133: bytes=32 time=39ms TTL=128
Reply from 101.68.8.133: bytes=32 time=42ms TTL=128
Reply from 101.68.8.133: bytes=32 time=44ms TTL=128
Reply from 101.68.8.133: bytes=32 time=61ms TTL=128
Reply from 101.68.8.133: bytes=32 time=15ms TTL=128
Reply from 101.68.8.133: bytes=32 time=30ms TTL=128
Reply from 101.68.8.133: bytes=32 time=33ms TTL=128
Reply from 101.68.8.133: bytes=32 time=41ms TTL=128
Reply from 101.68.8.133: bytes=32 time=28ms TTL=128
Reply from 101.68.8.133: bytes=32 time=38ms TTL=128
Reply from 101.68.8.133: bytes=32 time=39ms TTL=128
Reply from 101.68.8.133: bytes=32 time=51ms TTL=128
Reply from 101.68.8.133: bytes=32 time=16ms TTL=128

Ping statistics for 101.68.8.133:
    Packets: Sent = 4, Received = 13, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 15ms, Maximum = 61ms, Average = 36ms
```

Figure 81: ping from smartphone4 to smartphone6 in NET2.

Issues, and Limitations

In our project, we had some challenges, like working together as a team and understanding certain parts of the project. Even though it took time to figure out some parts, these challenges helped us get better at working as a team and solving problems. To fix these problems, we did extra research and talked about different ideas as a group. We also tried other ways to solve the problems to make sure we were doing things right. A significant issue we faced was in NET2 (Street Network), specifically with the backbone interface of the Central Office Server. While communication between the smartphones within NET2 (such as Smartphone4 and Smartphone5) worked successfully, we could not reach these devices from outside the network — for example, from PC0. After spending a lot of time troubleshooting, including reviewing tutorials and rechecking configurations, we discovered that the root cause was that the CO Server's backbone interface refused to accept an IP address. Every time we attempted to assign an IP, we received an error indicating an overlap or conflict, even though the subnet was calculated correctly. This made it impossible to properly route traffic between NET2 and the rest of the network, limiting connectivity and affecting overall functionality.

Teamwork

We worked together on every part of the project without dividing tasks. From IP subnetting and network design in Packet Tracer to configuring services, testing, and writing the report — we completed all steps collaboratively. We used Google Meet to communicate and coordinate our work, discussing each stage together, solving problems as a team, and making sure everyone was involved. This online teamwork helped us stay organized and maintain steady progress throughout the project.

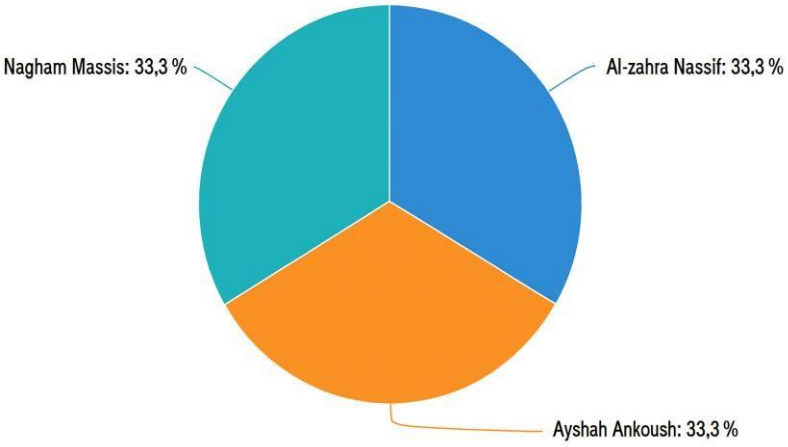


Figure 82: Teamwork Chart.

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